Briefing Session on 2022 HKDSE Geography Examination

22 & 23 November 2022

香港中學文憑 HKDSE
HONG KONG DIPLOMA OF SECONDARY EDUCATION

2022 HKDSE Geography
Report on Examination Statistics

	Year	2022	2021	2020	2019
No of candidates	SCH/1	**7 221**	7 566	8 042	8 263
	WG	**7 567**	7 957	8 381	8 631

2022 HKDSE Geography
Report on Examination Statistics

Level	2022	2021	2020	2019
5**	1.2	1.3	1.2	1.3
5*	4.9	5.1	4.9	5.1
5	12.5	12.9	11.9	12.3
4	37.4	39.1	38.3	38.5
3	62.8	65.8	64.7	64.0
2	88.0	88.6	88.2	87.5
1	97	97.5	97.5	97.3
U	3.0	2.5	2.5	2.7

Figures refers to Day Schools (SCH/1) %

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2022 HKDSE Geography
Report on Examination Statistics

Year	Mean (% mark)	SD (% mark)
2019	44.70	13.66
2020	45.93	13.59
2021	45.61	14.10
2022	**45.52**	**14.72**
Diff (2022-21)	**-0.09**	**0.62**

Figures refers to Day Schools (SCH/1) %

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2022 HKDSE Geography
Report on Examination Statistics

MCQ Statistics				
	2022	**2021**	**2020**	**2019**
Mean (% mark)	**11.2 (56.1)**	9.9 (49.3)	12.0 (60.1)	12.1 (66.2)
SD (% mark)	**2.9 (14.6)**	2.9 (14.3)	2.8 (14.1)	3.1 (15.3)
Lowest	**0**	1	1	0
Highest	**19**	19	20	20

Total no. of MCQ : 20

Figures refers to Day Schools (SCH/1) %

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2022 HKDSE Geography
Report on Examination Statistics

	Paper 1							
	Q1 (18)	**Q2 (18)**	**Q3 (18)**	**Q4 (18)**	**Q5 (18)**	**Q6 (12)**	**Q7 (12)**	**Q8 (12)**
Mean Score *(% marks)*	3.72 *(20.6)*	7.84 *(43.6)*	6.25 *(34.7)*	10.79 *(59.9)*	8.26 *(45.9)*	3.57 *(29.8)*	4.29 *(35.8)*	3.87 *(32.3)*
Highest Score	**14**	**18**	**16**	**18**	**18**	**11**	**11**	**12**
Popularity (%)	1	29	39	63	67	16	30	52

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Report on Examination Statistics

	Paper 2							
	Q1 (18)	**Q2 (18)**	**Q3 (18)**	**Q4 (18)**	**Q5 (12)**	**Q6 (12)**	**Q7 (12)**	**Q8 (12)**
Mean Score *(% marks)*	8.36 *(46.4)*	7.78 *(43.2)*	6.45 *(35.8)*	5.41 *(30.1)*	3.80 *(31.7)*	2.81 *(23.4)*	1.82 *(15.2)*	2.75 *(22.9)*
Highest Score	**18**	**18**	**14**	**15**	**11**	**10**	**10**	**11**
Popularity (%)	44	26	5	24	41	27	4	26

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School Survey on the 2022 HKDSE Geography Examination Papers

*The **duration** of the examination is appropriate.*

Strongly agree	Agree	Slightly agree	Slightly disagree	Disagree	Strongly disagree
4.6%	73.5%	17.2%	3.3%	0.7%	0.7%

*The level of **difficulty** of the examination is appropriate.*

Strongly agree	Agree	Slightly agree	Slightly disagree	Disagree	Strongly disagree
2%	74.7%	18%	3.3%	1.3%	0.7%

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School Survey on the 2022 HKDSE Geography Examination Papers

*The examination reflects the assessment **aims & objectives** stated in the **C & A Guide**.*

Strongly agree	Agree	Slightly agree	Slightly disagree	Disagree	Strongly disagree
6.8%	81.8%	8.8%	1.4%	1.4%	0%

*The examination is well-balanced in terms of **curriculum coverage**.*

Strongly agree	Agree	Slightly agree	Slightly disagree	Disagree	Strongly disagree
6.1%	80.3%	10.9%	1.4%	1.4%	0%

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School Survey on the 2022 HKDSE Geography Examination Papers

*An appropriate **balance** between questions testing **basic knowledge / skills & those testing higher-order skills.***

Strongly agree	Agree	Slightly agree	Slightly disagree	Disagree	Strongly disagree
2.7%	79.2%	14.8%	1.3%	1.3%	0.7%

*The **wording** of the questions is clear & easily understandable.*

Strongly agree	Agree	Slightly agree	Slightly disagree	Disagree	Strongly disagree
5.4%	83.1%	8.8%	2%	0.7%	0%

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School Survey on the 2022 HKDSE Geography Examination Papers

*The **optional questions** are at a* comparable ***level of difficulty**.*

Strongly agree	Agree	Slightly agree	Slightly disagree	Disagree	Strongly disagree	N.A.
1.4%	60.1%	25%	4.7%	3.4%	1.4%	4.1%

*The **two language versions** are identical in meaning.*

Strongly agree	Agree	Slightly agree	Slightly disagree	Disagree	Strongly disagree	N.A.
3.3%	68.7%	6.7%	0.7%	0%	0%	20.7%

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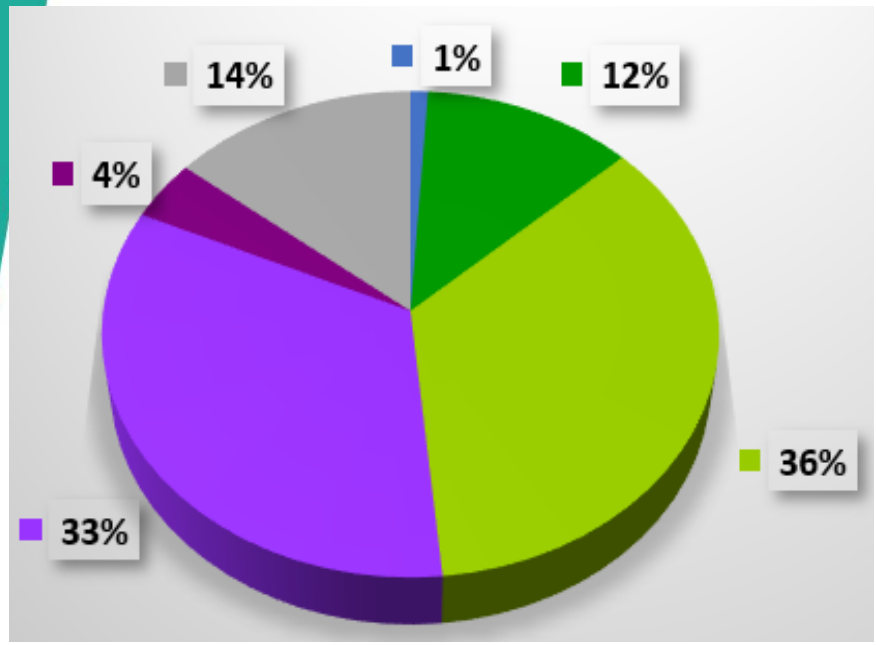
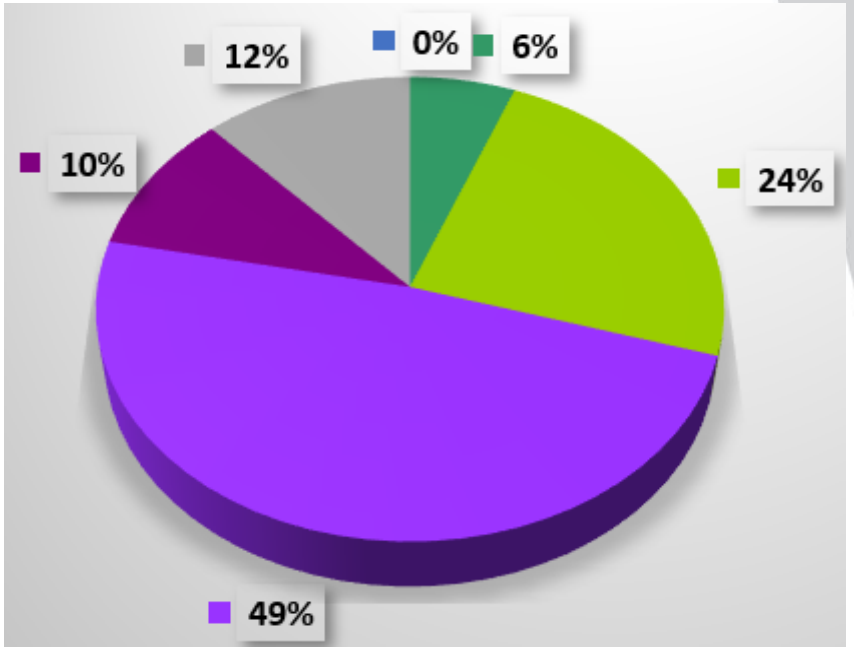
Comments from Markers on Candidates' Standard Compared with 2021 Examination

Paper 1



Paper 2



much better
Slightly better
about the same
slightly poorer
much poorer
N.A.

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2022 HKDSE Geography Examination

Review of Candidates' Performance in Paper 1

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General comments and recommendations

- The overall performance of candidates was about the same as last year.

- Candidates commonly demonstrated inadequate knowledge and understanding of the geographical issues and concepts in answering questions that require explanations and logical reasoning. They should not depend on rote memorisation as their primary means of learning.

- Candidates should take note of the key concepts and learning points of each unit. They should also incorporate the key geographical concepts of location and distribution in learning so as to understand the geographical phenomenon, processes and issues etc. with a spatial perspective.

- Candidates should be aware of the changing geographical settings and phenomena, and the development of geographical issues with time, and update their knowledge and information continuously.

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General comments and recommendations (Cont')

- Candidates should interpret questions and information provided carefully, and respond to the questions appropriately. They should also apply appropriately their knowledge to the specific situations stated in the questions.

- Candidates should pay attention to the balance of breadth and depth of content in answering questions and give appropriate elaboration for the key points. They should also learn how to cite examples to illustrate their answers.

- Candidates commonly were able to discuss from different perspectives and showed a clear stance in answering questions with open-ended discussion. However, quite a large number of candidates used 'someone said/ thought that...' or 'there are opinions that...' etc. so as to avoid being taken as 'contradictory arguments' by the markers. They should have their own perspective for discussion.

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Section A

- The average number of questions answered correctly by candidates was 11 out of 20 (10 in 2021) multiple-choice questions.
- The overall performance of candidates was about the same as last year.
- Four questions in which distractors were more popular than the key have been selected for further discussion.

Section A (Cont')

QT 11 **Plant X is still a major iron and steel plant in China because _____.**

(1) there is agglomeration of heavy industries
(2) its production facilities have been upgraded
(3) there is abundant cheap labour supply

* A.	(1) and (2) only	(31%)
B.	(1) and (3) only	(32%)
C.	(2) and (3) only	(8%)
D	(1), (2) and (3)	(29%)

In Item 11, the most popular answer was Option B.

Candidates choosing this option might have mistaken iron and steel industry in China as being labour intensive.

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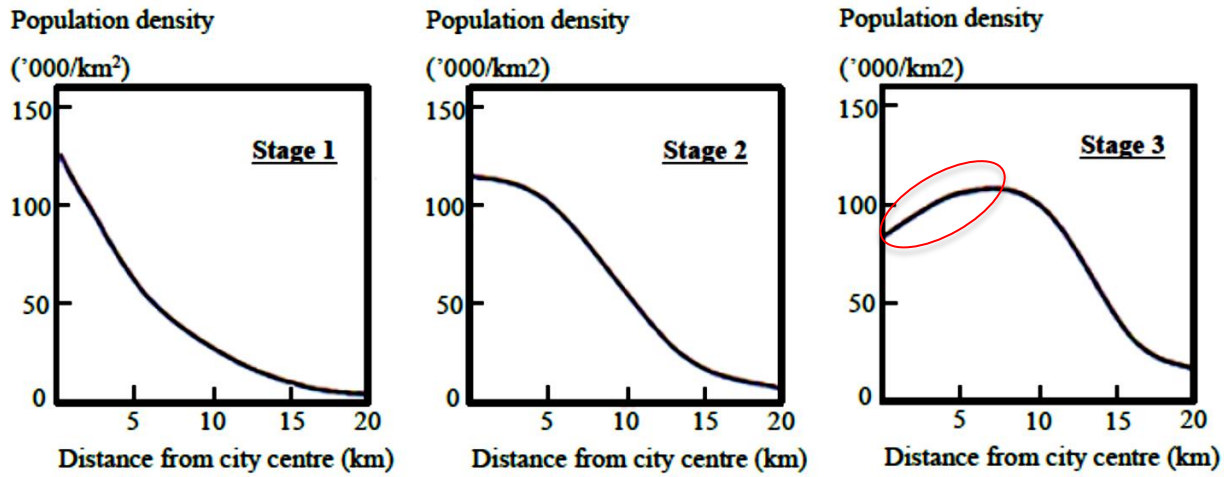
Section A (Cont')

Q.13 The graphs below show the changes in population density at different stages of development of a city.



Which of the following statements is/ are correct?

(1) Population density decreases with the distance from the city centre at the different stages of development.
(2) More people move back to the city centre at the later stage of development.
(3) The city experiences urban growth.

A.	(1) only	(42%)
* B.	(3) only	(26%)
C.	(1) and (2) only	(17%)
D	(2) and (3) only	(15%)

In Item 13, the most popular answer was Option A.

Candidates choosing this option might have misinterpreted the graph at stage 3.

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Section A (Cont')

In Item 16, the most popular answer was Option A.

Candidates choosing this option might be unaware of the rising contaminated underground water towards the surface during the hot dry summer in Southern California.

Q.16 Which of the following are the possible consequences of inappropriate use of chemical fertilizer in Southern California?

(1) underground water pollution
(2) eutrophication
(3) salinization of soil

A.	(1) and (2) only	(52%)
B.	(1) and (3) only	(9%)
C.	(2) and (3) only	(11%)
* D	(1), (2) and (3)	(28%)

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Section A (Cont')

In Item 17, the most popular answer was Option D.
Candidates choosing this option might have mistaken urban development, which cleared a much smaller area of Amazon forest when compared with economic development, as a 'major' cause.

Q.17 What are the major causes of the high rate of deforestation in the rainforest in the Brazilian Amazon?

(1) high population growth
(2) poverty
(3) rapid urbanisation

* A.	(1) and (2) only	(29%)
B.	(1) and (3) only	(16%)
C.	(2) and (3) only	(7%)
D	(1), (2) and (3)	(48%)

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Section B Question 1 (FBQ)

	Area	Key concepts/ skills
(a)	Factors to be considered in field work planning	• Validity, reliability of field work • Risk assessment
(b) (i)	Fieldwork skills – data collection	• Use of sampling method (Systematic sampling) • Use of equipment/ tools for data collection
(ii)	Fieldwork skills – data processing, presentation and analysis	• Application of knowledge and concepts in channel study: depth, width, calculation of cross-sectional area and discharge • Statistical techniques: e.g. mean • Use of analytical methods: e.g. diagrams for data analysis
(c)	Evaluation of fieldwork	• Critical evaluation • Pros and cons of the data collection methods: e.g. sampling, sites selection, instruments used etc. • Reliability and validity of the study (both in evaluation and suggestions)

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Section B Question 1 (FBQ) (Cont')

(a) Refer to Table 1a. State one merit and one demerit of the date for conducting the fieldwork. Explain your answer.

Fieldwork topic	The relationship between river courses and **river discharge**
Fieldwork date	A day in July

✓ Most candidates stated the potential risks in conducting fieldwork in the rainy season as the demerit

× Quite a large number of candidates failed to state the merit.

× Most candidates failed to explain the merit and demerit in terms of the validity and reliability of data collected.

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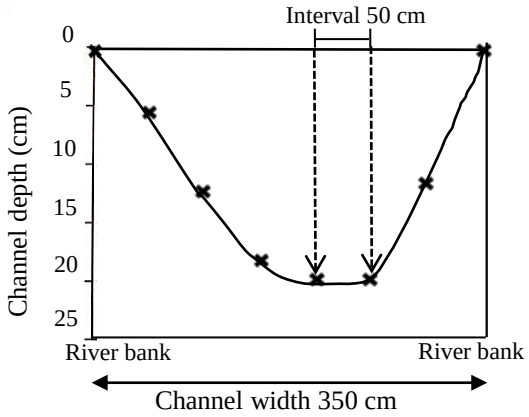
Section B Question 1 (FBQ) (Cont')

(b) Refer to Tables 1a and 1b.

(i) Describe how to use the tools listed in Table 1a to collect the data of channel depth and width in order to draw the river channel cross section.

Tools for collecting the data of channel depth and width	A measuring tape, a ruler and two ranging poles



✓ Many candidates described how to use the measuring tape and ruler to measure the channel width and depth respectively.

× Most candidates failed to describe the proper use of the ranging poles.

Section B Question 1 (FBQ) (Cont')

(b) (ii) The students compiled a fieldwork report with the hypothesis: '**River discharge** increases downstream'.
How should the students process and utilise the data collected before hypothesis testing?

		Site for data collection			
		A	B	C	D
Velocity (m/s)	Record 1	0.18	0.20	0.68	0.89
	Record 2	0.23	0.23	0.12	0.92

× Most candidates failed to describe how to utilise the data processed to test the hypothesis.

× Candidates were commonly unfamiliar with the data processing procedure and the statistical techniques required. They failed to describe clearly the steps how to calculate the river discharge from the data collected.

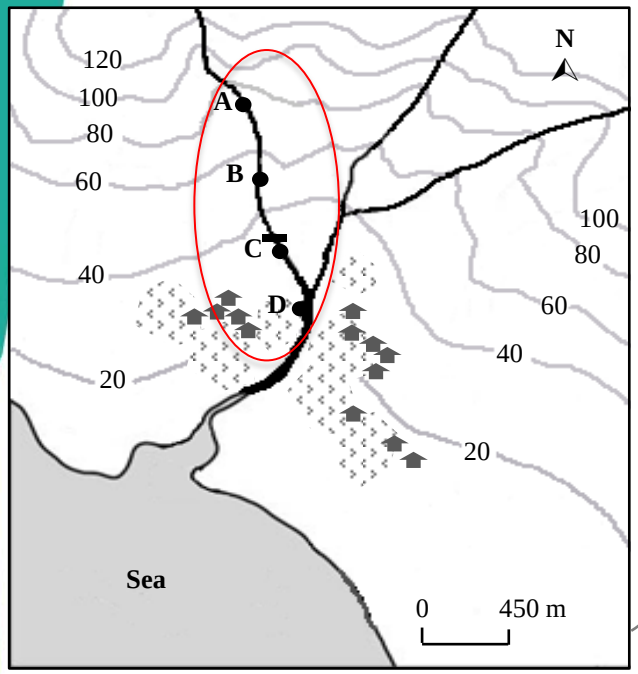
× Some candidates did not realise the needs to calculate the 'mean' value of the data.

× Many candidates failed to respond appropriately to the question. They described the drawbacks of data collection methods or the problems of using extreme data.

Section B Question 1 (FBQ) (Cont')

(c) Refer to Tables 1a and 1b.

Evaluate the data collection methods used in this fieldwork and suggest ways for further improvement.



Key:
- Site for data collection
- Farmland
- Settlement
- Weir
- River
- 20 Contour line (V.I. 20 metres)

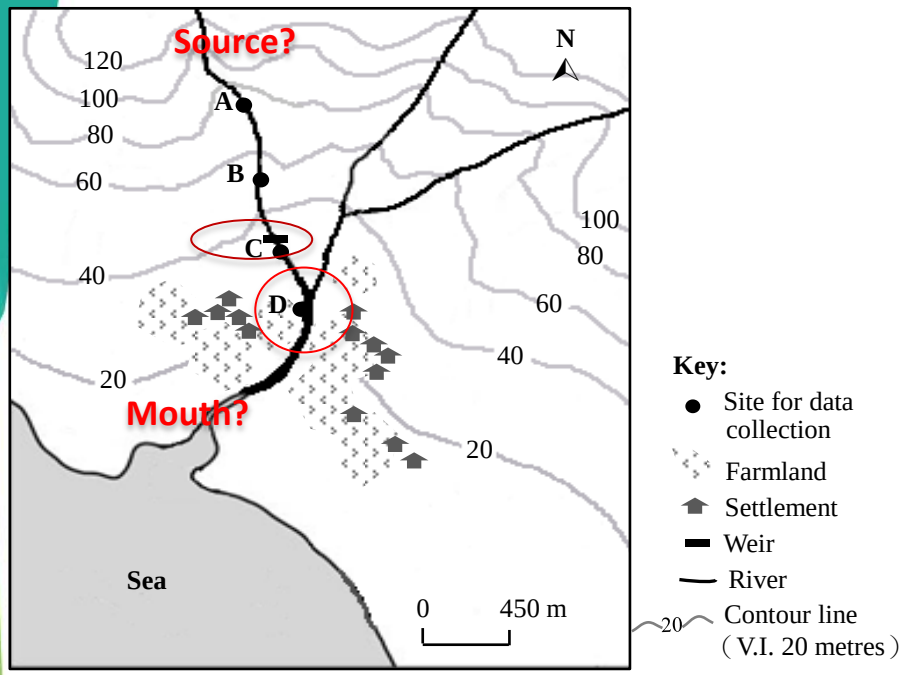
		Site for data collection			
		A	B	C	D
Velocity (m/s)	Record 1	0.18	0.20	0.68	0.89
	Record 2	0.23	0.23	0.12	0.92

× Candidates commonly failed to evaluate the data collection methods satisfactorily due to their poor understanding of the key concepts of validity and reliability in evaluation of the fieldwork.

× Most candidates only stated briefly some general drawbacks of data collection, such as an inadequate number of sites and records, and failed to explain further.

Section B Question 1 (FBQ) (Cont')

Use **systematic sampling** to select 4 sites, A, B, C and D with an interval of approximately 450 m to collect data of the river channel



× Few candidates made a comprehensive evaluation of the data collection methods used.

× Candidates commonly failed to comment on the sampling methods used and the appropriateness of sites chosen.

× Most of candidates' suggestions for improvement were also brief and general. These included choosing more sites for data collection or taking more records at each site etc. Candidates failed to explain further how these suggestions might improve the validity and reliability of the fieldwork.

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Section B Question 2

	Area	Key concepts/ skills
(a)(i)	Coastal processes	• Interpretation of aerial photo and photograph • Erosion
(ii)	Factors influencing coastal erosion process	• Factors affecting wave energy: location, relief, fetch, offshore gradient etc. • Wave refraction • Data interpretation
(b)(i)	Coastal processes and formation of landform features	• Wind direction, deposition, longshore drift • Spit and tombolo • Diagram drawing and annotation • Interpretation of aerial photo
(ii)	Human activities influencing coastal environment	• People – environment interaction • Reclamation, raised coastal land, coastal management (hard management strategies) • Interpretation of aerial photo
(c)	Coastal management issues: evaluation of the strategies and their possible impacts	• People – environment interaction, conflicts • Cost – effectiveness of management strategies • Sustainable management of environment • Logical discussion and decision making • Interpretation of aerial photo

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Section B Question 2 (Cont')

(a) Refer to Figures 2a and 2b.

(i) Identify the major coastal process in area P.

Area P

✓ About three quarters of candidates correctly identified the major coastal process in P as erosion.

× The other candidates either wrongly identified it as deposition or gave irrelevant answers such as type of wave, name of coastal feature etc.

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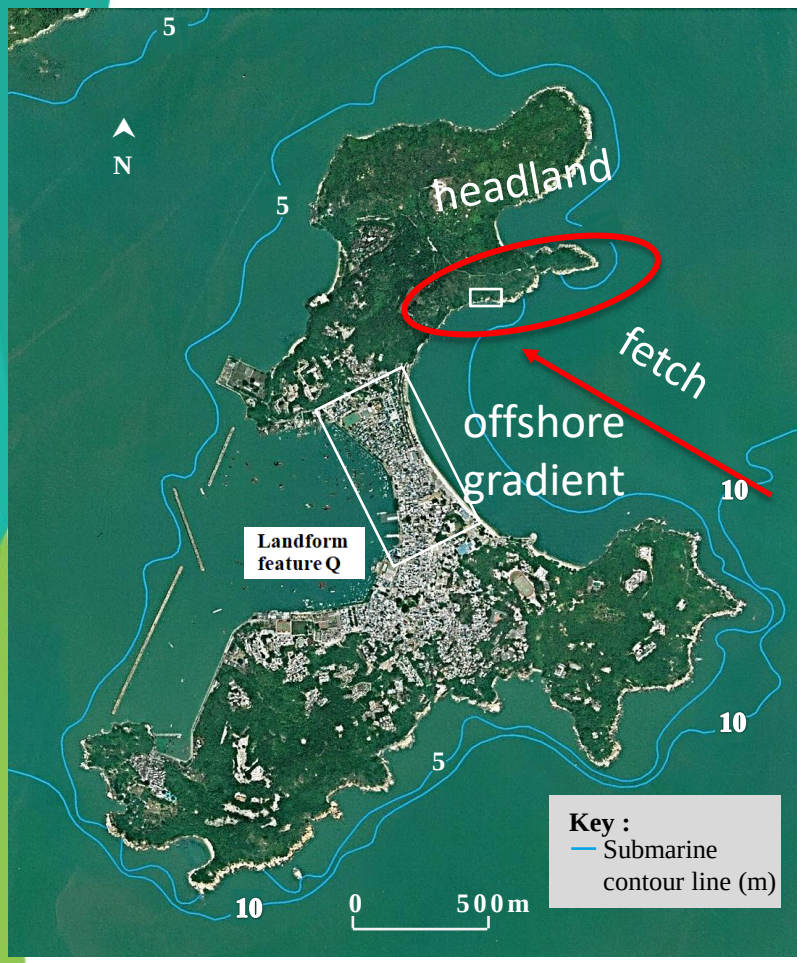
Section B Question 2 (Cont')

(ii) Explain how the physical settings of area P favour the above major coastal process.



✓ Most candidates were able to state headland location, long fetch and steep offshore gradient etc. as the favourable physical settings for coastal erosion.

× Some candidates confused 'offshore gradient' with 'depth of coastal water'.

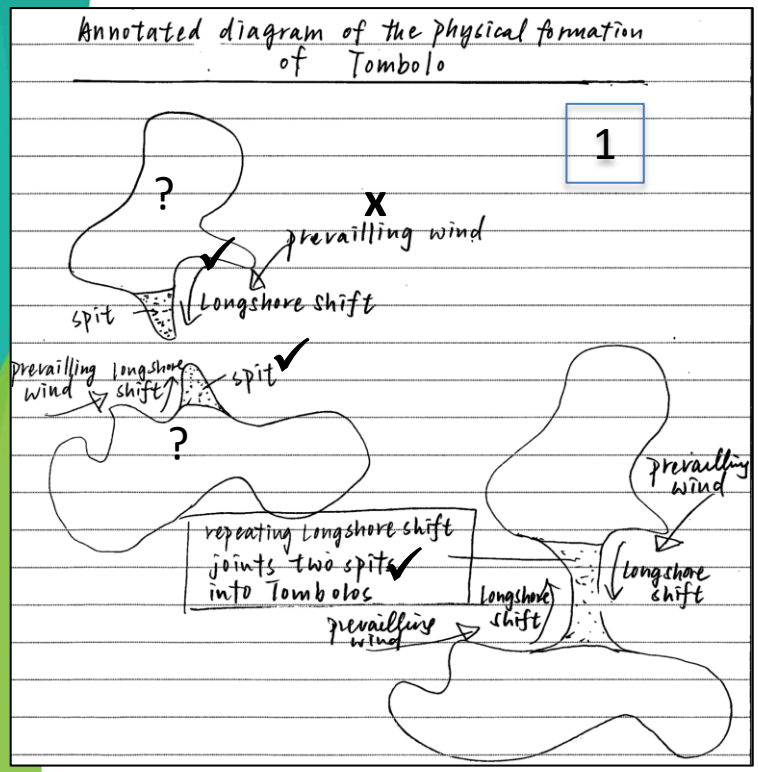
× Candidates commonly did not have the concept of 'energy' of breaking waves, thus they failed to explain clearly how these physical settings favour coastal erosion.

× Many candidates mentioned 'destructive waves' in their answers, but did not explain adequately how it is related to the physical settings at P.

Section B Question 2 (Cont')

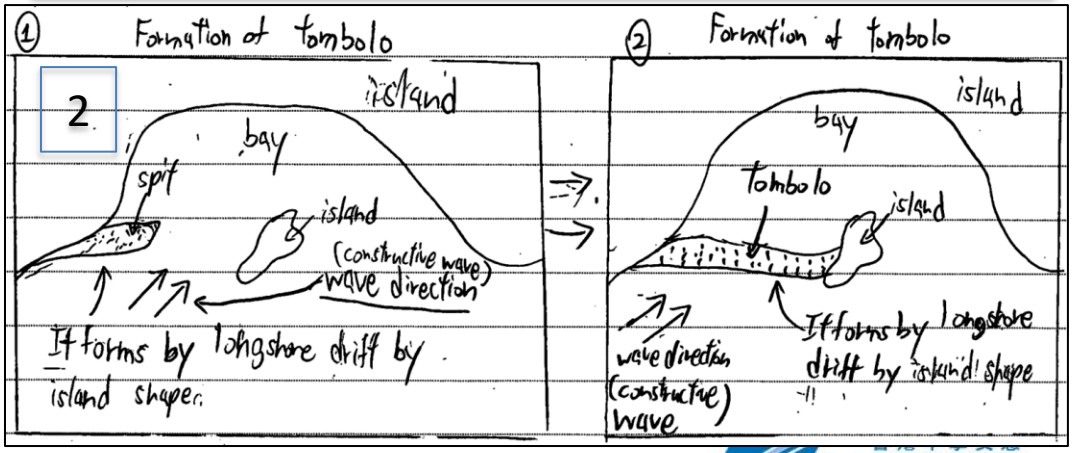
(b) Refer to Figure 2b and Photograph 2c.

(i) Identify landform feature Q. Draw a series of annotated diagrams to explain its physical formation.



✓ Most candidates correctly identified landform feature Q as tombolo, and were commonly able to draw diagrams showing the development stages from growing of spits to connecting of the islands.

× Many diagrams were not well drawn.



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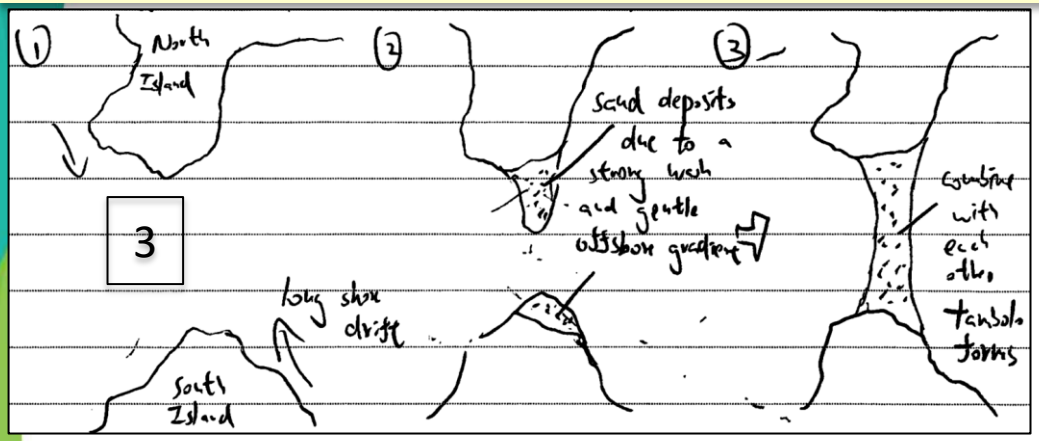
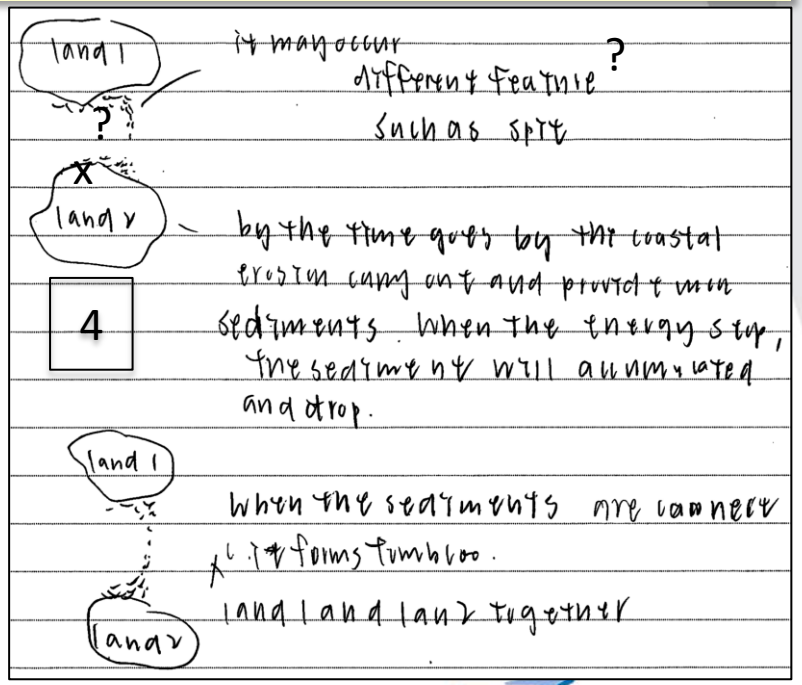
Section B Question 2 (Cont')

- × Many candidates failed to show the key factors of tombolo's formation, i.e. directions of wind and longshore drift, in their drawings.
- × Quite a number of candidates drew the directions of monsoon wind inaccurately.
- × Some candidates erroneously thought that the tombolo of Cheung Chau was formed by the extension of one spit only; some others inaccurately labelled 'spit' as a 'sand bar'.



- × Some candidates did not appropriately annotate their diagrams and wrote lengthy explanations next to the diagrams.



Section B Question 2 (Cont')

(ii) Account for the human modifications of landform feature Q for development.

- ✓ Most candidates identified reclamation, sea wall and other man-made structures on landform feature Q.
- ✓ Some of them gave appropriate explanations in terms of the man-land inter-relationship in Cheung Chau.
- ✓ A few candidates noted the raising of the tombolo to avoid being submerged at high tides.

- × Quite a number of candidates were seemingly unaware of the key word 'modification' in the question, they failed to state clearly how landform Q was 'modified'.
- × Some candidates accounted for the construction of breakwater which is offshore.

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Section B Question 2 (Cont')

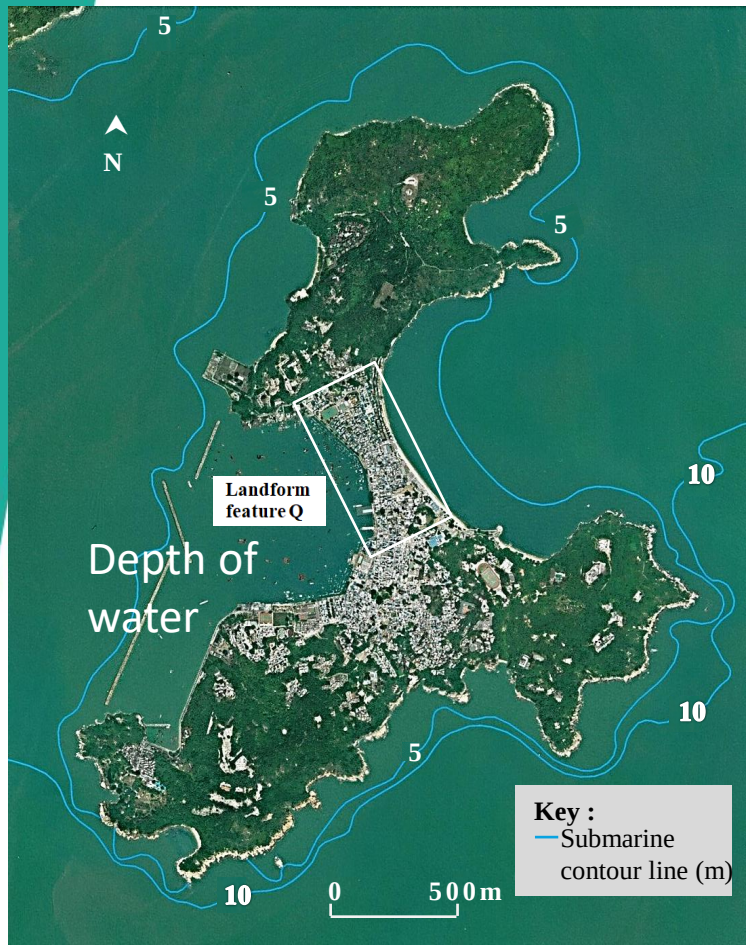
(iii) *Some hard coastal management strategies have been adopted on Cheung Chau.*

Discuss whether they are appropriate strategies to manage both the eastern and western coasts of landform feature Q.

- ✓ Most candidates identified breakwater and described its use to protect the typhoon shelter.
- ✓ Some candidates identified sea wall.

- × Candidates' discussions were commonly insufficient to justify whether these strategies were appropriate or not.
- × Many candidates limited themselves to putting forward the general pros and cons, limitations and drawbacks of the strategies, such as 'protect the coast', 'reduce erosion' or 'high cost' etc. as their arguments

Section B Question 2 (Cont')



× Candidates' discussions commonly focused on the western coast only.

× Few candidates compared the differences in the physical and human environments between the eastern and western coasts, and discussed from the perspective of man-land interrelationship.

× Some candidates mentioned 'beach nourishment' but it is irrelevant.

Section B Question 3

	Area	Key concepts/ skills
(a) (i)	Urban problems	• Housing, transport problems (decay, environmental conditions etc.) • Photograph interpretation
(ii)	Processes involved in urban growth	• Change over time: population growth, change in urban structure and infrastructure • Location, urban decay process • Photograph and map interpretation (1:20 000)
(b)(i)	Internal structure of a city (new town)	• Urban land uses, spatial distribution • Map interpretation (1:20 000)
(ii)	Urban planning (land use zoning) and sustainable development	• People – environment interrelationship • Characteristics of a sustainable city: Socio-economic, environment • Map interpretation (1:20 000)
(c)	Measures to alleviate urban problems and conflicts arising / costs	• People – environment interrelationship • Comprehensive urban renewal (social, economic and environmental costs/ conflicts) • Map interpretation (1:20 000) • Logical discussion and decision making

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Section B Question 3 (cont')

(a) Refer to Figure 3 and the map extract (1:20 000) of Hong Kong.

(i) What are the urban problems in area X as shown in Figure 3?

✓ Most candidates identified air pollution, traffic congestion etc. as the urban problems in area X.

× Many candidates inaccurately stated 'old buildings' instead of 'derelict buildings' as a problem.

× Many candidates were unable to differentiate between causes and problems. High building density, mixed land use etc. are causes, but not urban problems.

× Some candidates misinterpreted the question and accounted for the occurrence of urban problems.

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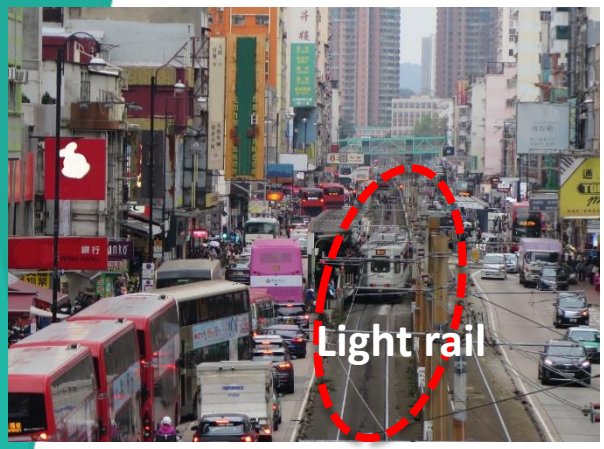
Section B Question 3 (Cont')

(ii) Account for the occurrence of the urban problems mentioned in (a) (i).



✓ Most candidates mentioned 'outdated urban planning' and 'land use conflict' as the causes leading to the occurrence of urban problems.

× Candidates commonly demonstrated inadequate understanding of these two factors.

× Many candidates failed to use appropriate local examples to illustrate 'land use conflict'.

× Candidates commonly did not realise that mixed residential and commercial land use does not always cause land use 'conflict' problems.

× Some candidates mentioned some problems that are caused by poor management rather than deficiencies in planning.

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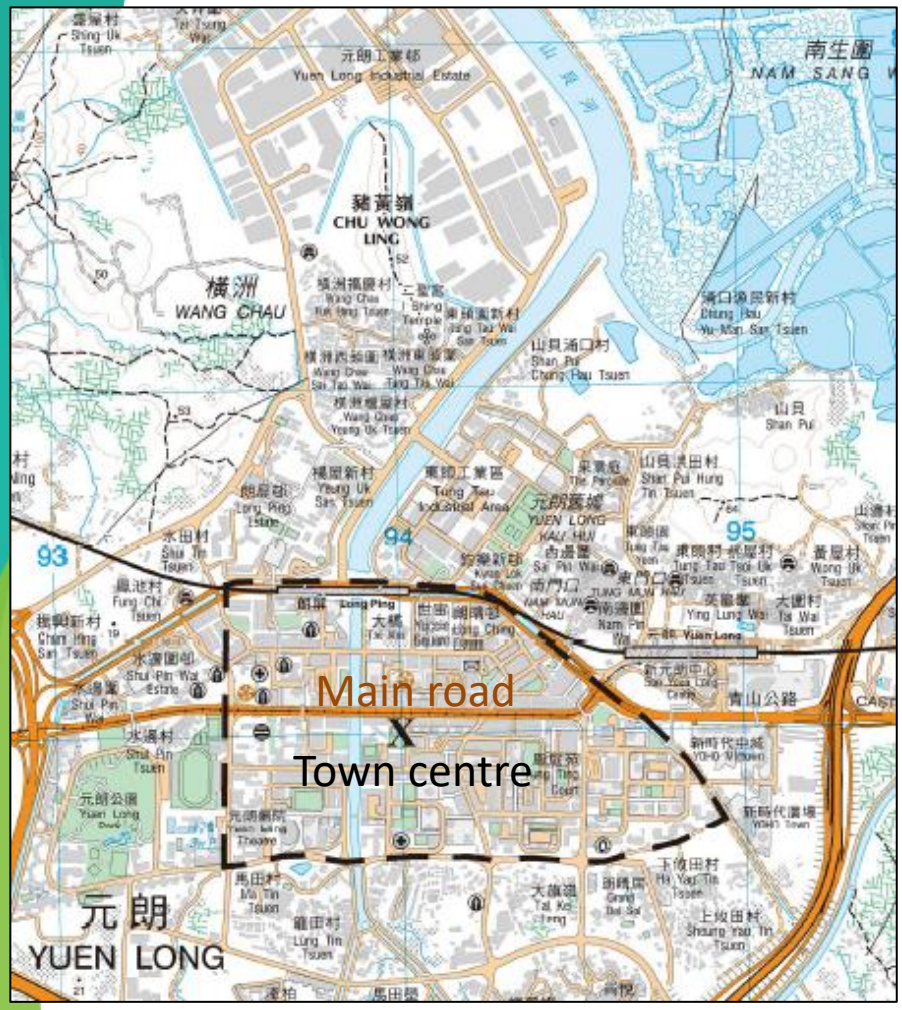
Section B Question 3 (Cont')

(ii) Account for the occurrence of the urban problems mentioned in (a) (i).



× Most candidates did not refer to the map extract.

× Without an overall picture of the spatial distribution of land uses and transport network in the area, they failed to explain specifically, from a broad perspective, how the deficiencies in land use zoning and transport planning might cause the occurrence of urban problems.

× Many candidates only gave some superficial and general explanations, not specifically related to the problems they mentioned in (a)(i).

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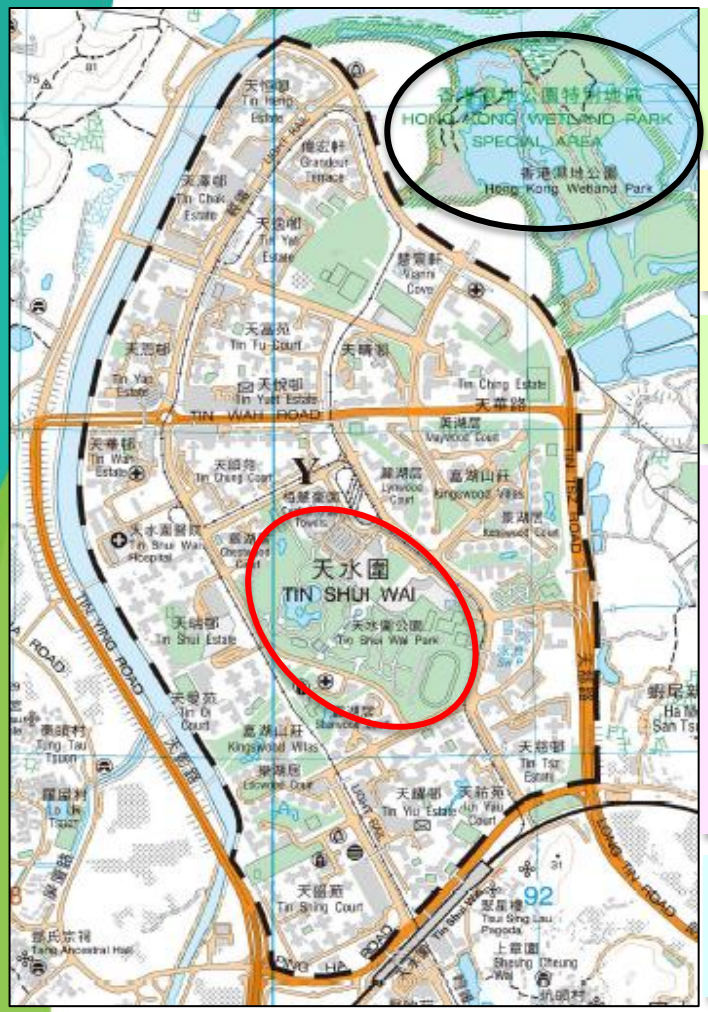
Section B Question 3 (Cont')

(b) Land use zoning is implemented in area Y. Refer to the map extract (1:20 000) of Hong Kong.

(i) Describe the spatial distribution of residential land use and recreational land use in area Y



✓ Most candidates were able to identify the residential and recreational land uses in area Y.

× Some candidates confused 'institutional' with 'recreational' land use.

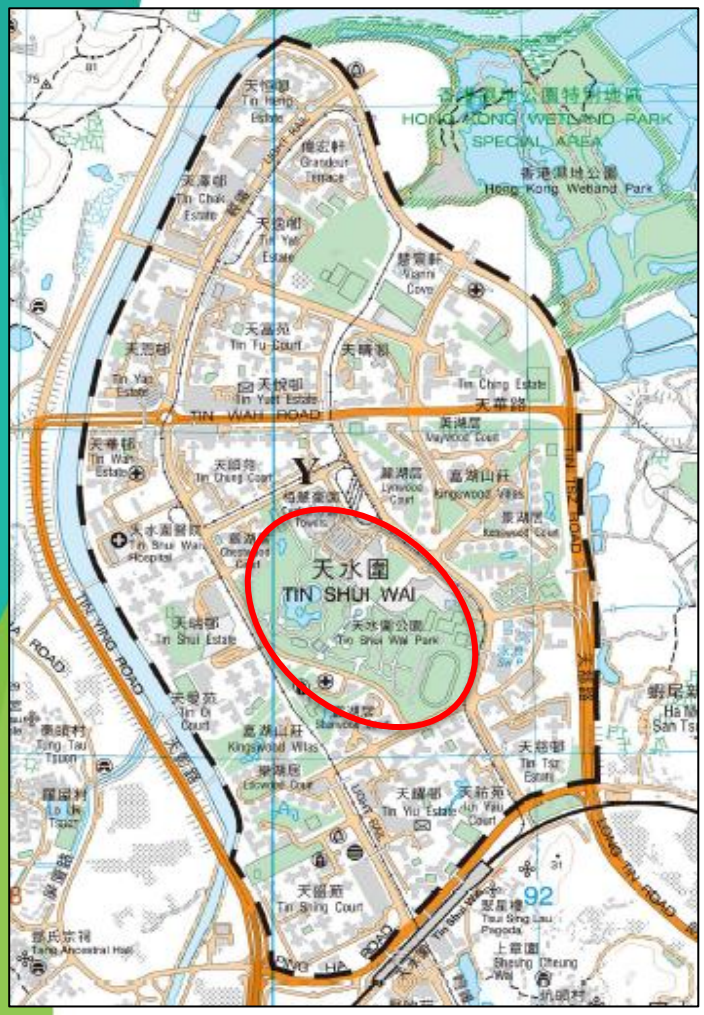
✓ Many candidates described accurately the central location of Tin Shui Wai Park.

× Quite a number of candidates demonstrated an inadequate understanding of geographical terminology used in spatial distribution. They used inappropriate descriptions such as abundant, average or evenly distributed etc. to describe the scatter distribution of recreational land use.

× Some candidates inappropriately cited examples outside the area Y to illustrate, such as Wetland Park.

Section B Question 3 (Cont')

(ii) What are the merits of the above land use distribution in area Y?



× Candidates commonly lacked in-depth understanding of land use zoning in Hong Kong's new town.

× Only a few candidates identified the specific merits of land use 'zoning' in Y, such as the use of park as 'buffer zone'.

× Most candidates failed to refer to the spatial distribution of the land use, but listed out some general merits of abundant recreational land use in area Y only, such as 'more open space', 'better living quality' etc.

× Some candidates used inappropriate terms 'green belt' and 'living standard' in their answers, some others confused the concept of 'housing density' with 'population density'.

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Section B Question 3 (Cont')

(c) Refer to Figure 3 and the map extract (1:20 000) of Hong Kong.

With reference to the land use zoning in area Y, discuss whether it is an appropriate strategy to tackle the urban problems in area X.

× Candidates commonly failed to discuss specifically whether the land use zoning might tackle the urban problems in area X or not. Supporting arguments for adopting land use zoning were commonly repeating the merits mentioned in (b)(ii), without stating how the urban problems e.g. traffic congestion, air and noise pollution etc. be tackled.

× Some candidates inappropriately judged merely by the size of area X and Y.

× Candidates commonly realised that re-zoning involved urban renewal. The arguments against zoning were commonly only some general problems and difficulties encountered in urban renewal.

× Few candidates referred to the specific situation of area X, and discussed the challenges to implement a large scale comprehensive urban renewal in a highly accessible and busy commercial and residential area.

× Some candidates raised irrelevant arguments such as gentrification after renewal, air and noise pollution in renewal process.

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Section B Question 4

	Area	Key concepts/ skills
(a) (i)	Structure of tropical rainforest – emergent layer	• Structure of TRF • Links and interrelationships between abiotic and biotic components (trees of emergent layer) • Data interpretation
(ii)	Structure of tropical rainforest – undergrowth layer, lianas	• Structure of TRF • Links and interrelationships between abiotic and biotic components (lianas and decomposers) • Photograph interpretation
(b)	Impact of large scale deforestation on vegetation structure and microclimate	• Impact of large scale deforestation • Links and interrelationships between abiotic (microclimate) and biotic components (vegetation) • Data interpretation
(c)	Measures to protect TRF - afforestation	• Evaluation of conservation and management measures of TRF • Effectiveness in restoration of the ecosystem (density, biodiversity, succession process etc.) • Problems of protecting TRF (socio-economic costs) • Data interpretation • Logical discussion and decision making

Section B Question 4 (Cont')

(a) Refer to Figure 4a.

(i) Describe and explain the characteristics of the trees in layer X.

✓ Most candidates described the characteristics of trees in the emergent layer.

× Some candidates lacked the key concept of 'keen competition among plants' in their explanations.

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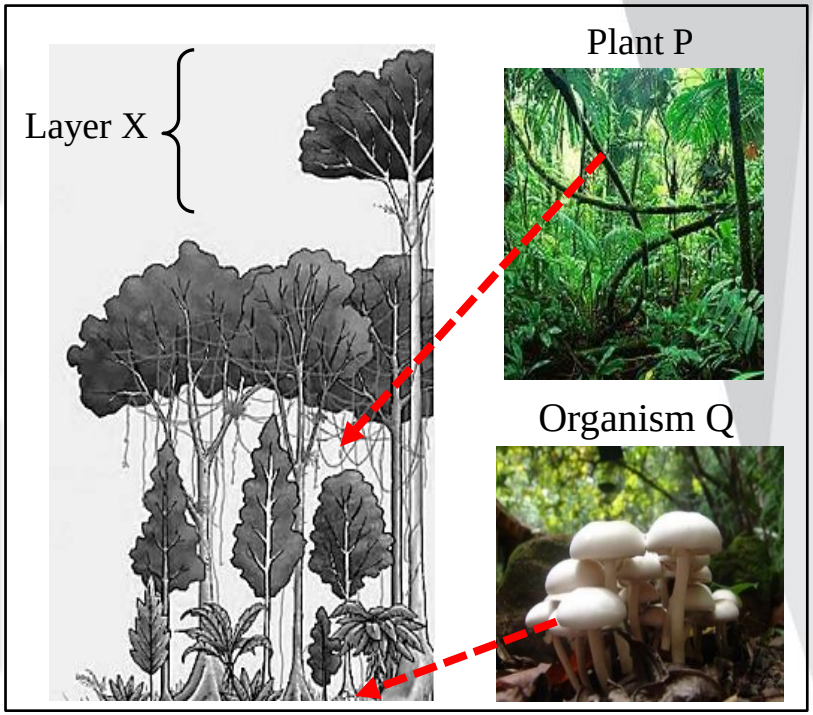
Section B Question 4 (Cont')

(ii) With reference to the specific environmental conditions of different layers, account for the growth of plant P and organism Q.



✓ Many candidates were able to identify correctly organism Q.

× Quite a large number of candidates inaccurately identified plant P as epiphytes or stranglers.

✓ Candidates were commonly able to describe the characteristics of plant P and organism Q, as well as the environmental conditions of canopy layer and undergrowth layer.

× Many candidates failed to explain specifically the growth of climber and fungi with reference to their inter-relationships with the biotic and abiotic environments of the layers, such as 'competition for sunlight' or 'abundant litter available'.

× Quite a number of candidates inaccurately explained the growth of climbers or fungi as parasitic.

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Section B Question 4 (Cont')

(b) Refer to Table 4b. Explain how the changes in vegetation structure after clear-cutting affect the microclimate in a tropical rainforest.

	Before clear-cutting	After clear-cutting
Temperature (ºC)	24.4	29.9
Light intensity (lux)	3 856	20 960
Relative humidity (%)	97.2	92.3
Wind speed (m/s)	0.3	2.9

✓ Candidates were commonly able to explain the changes in microclimate after clearing of rainforest

× Some candidate confused microclimate with global warming, they inaccurately applied the concept of 'carbon sink' to account for the temperature increase.

× Some candidates confused 'evaporation' with 'transpiration'.
× Some others confused cause with effect, they wrongly stated 'increase in evaporation' as the cause of decrease in relative humidity.

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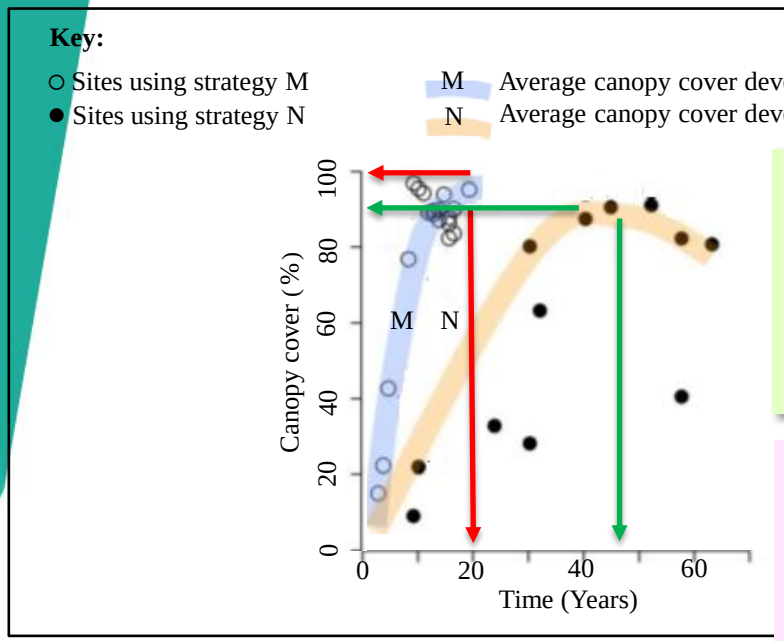
Section B Question 4 (Cont')

(c) Refer to Figure 4c. Discuss whether strategy M or N is more preferable in restoring a cleared tropical rainforest area.



✓ Most candidates referred to the graph and put forward simple and straight forward arguments, such as 'faster development' or 'higher cover rate' etc.

× Candidates commonly failed to discuss further in terms of the 'effectiveness' or 'reliability' of the strategies, or the importance of canopy cover to the restoration of rainforest

× Some candidates misinterpreted information and put forward some unsound arguments, such as the canopy will disappear suddenly after a certain number of years.

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Section B Question 4 (Cont')

Strategy M	Strategy N
● Intensive human intervention ● Well-designed tree planting with high density and diversity	● Limited human intervention ● Natural regrowth of vegetation

× Candidates commonly failed to analyse other information and also failed to apply their knowledge effectively.

× Only a few candidates explained specifically the importance of the high diversity of trees to rainforest restoration.

× Many candidates argued that 'intensive human intervention' is undesirable but their arguments were commonly general or irrelevant e.g. 'not natural', 'using chemicals' or 'building roads' etc. Candidates seemingly had a limited understanding of the positive role of man in rainforest management.

× They also apparently did not realise that most rainforest countries are less developed countries with limited human inputs.

× Some candidates put forward some unsound arguments, such as natural regrowth will cause low diversity of a rainforest, or strategy M is planting for timber or ecotour etc.

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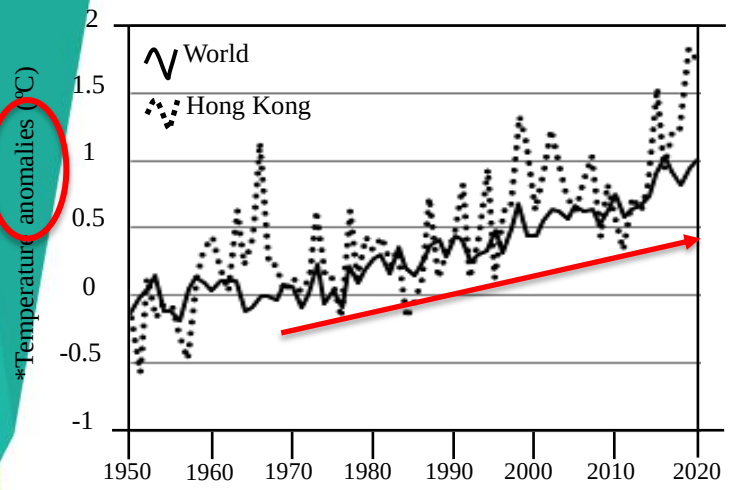
Section B Question 5

	Area	Key concepts/ skills
(a)	Natural and human causes of global warming	• The mechanism of greenhouse effect • The role of human activities in enhancing greenhouse effect with reference to global economic development • Graph interpretation
(b)(i)	Climate change at local scale	• Observed climate change in HK • Graph interpretation
(ii)	Impact of urban growth and development on local climate (temperature)	• Man - environment interrelationship • Urban growth and development: urban activities, urban structure and land uses • Urban climate (atmospheric composition, humidity, wind etc.) and heat island effect • Photograph interpretation
(c)	Impact of climate change and mitigation measures at local level	• Man-environment interrelationship • Consequence of climate change e.g. health risks, extreme weather events etc. • Mitigation measure: environmental friendly building designs and local climate • Sustainability, cost - effectiveness of the measure • photograph interpretation • Logical discussion and decision making

Section B Question 5 (Cont')

(a) Refer to Figure 5a. Describe and explain the overall trend of temperature changes of the world from 1950 to 2020.



✓ Candidates commonly described correctly the trend of an increasing world temperature.

× Many candidates seemingly did not realise the differences between 'temperature' and 'temperature anomalies'. Hence they inaccurately used 'temperature anomalies' in answering.

✓ Most candidates understood the relationship of greenhouse gases and global warming.

× Many candidates failed to respond appropriately to the 'temperature changes from 1950 to 2020'. They either listed out various human sources of greenhouse gases, or inappropriately cited 'Industrial Revolution' in their explanation.

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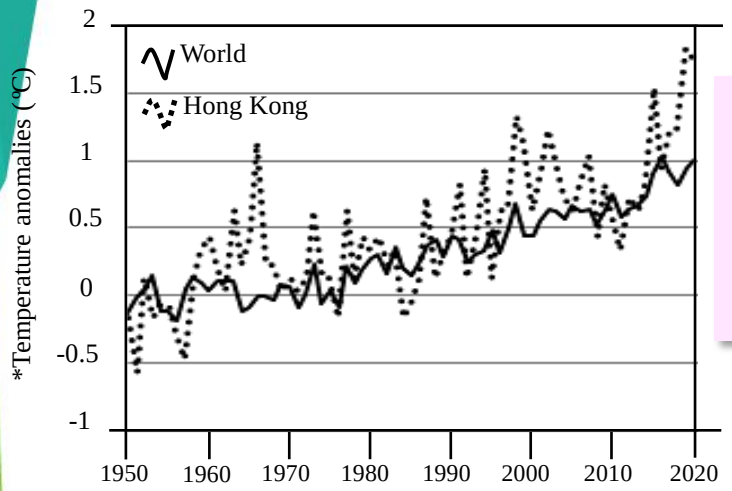
Section B Question 5 (Cont')

✓ Candidates commonly demonstrated basic understanding of the mechanism of greenhouse effect.

× However, many candidates lacked the concept of 'greenhouse gases absorb terrestrial radiation', thus they also failed to explain 'counter radiation'.

× Some candidates confused the concepts of 'radiation' with 'reflection' in answering.



× Some candidates did not study the question carefully, they either described changes of world temperature in great detail or compared it with the temperature changes in Hong Kong.

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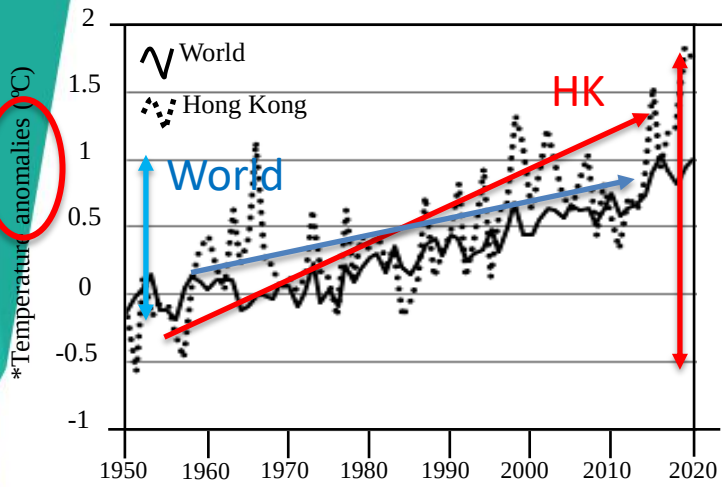
Section B Question 5 (Cont')

(b) Refer to Figures 5a and 5b.

(i) How is the magnitude of temperature changes in Hong Kong different from that of the world?



✓ Most candidates were able to state the difference in the magnitude of temperature changes.

× Many candidates inaccurately used the term 'temperature anomalies' in answering.

Section B Question 5 (Cont')

(ii) Explain the above difference.

✓ Candidates commonly were able to apply the concept of 'heat island effect' to explain.

✓ Most candidates explained well the 'wall effect' of dense and tall buildings.

× Many candidates may have confused 'greenhouse effect' with 'heat island effect'. They applied the concept of 'carbon sink' to explain the impact of 'less vegetation cover in the urban areas'.

× Many candidates used 'increase in greenhouse gases', instead of 'waste heat' generated in their explanation of 'usage of vehicles and air conditioners'.

× Many Some candidates wrongly applied the concept of 'blanketing effect' of cloud.

× Some candidates did not refer to Figure 5b and cited inappropriately the development of manufacturing, transport and logistics in Hong Kong in the last several decades in their explanation.

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Section B Question 5 (Cont')

(c) Refer to Figures 5a, 5b and 5c.

Discuss whether Hong Kong should adopt the building designs shown in Figure 5c as a measure to tackle local temperature changes.

✓ A few candidates explained correctly the cooling effect of greenery to the building and thus reducing use of air conditioners and waste heat generated.

× Candidates commonly demonstrated poor understanding of 'urban greening'. Many candidates inappropriately explained the role of plants is to absorb carbon dioxide and pollutants.

× Some candidates inaccurately used the term 'green belt' for the greenery.

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Section B Question 5 (Cont')

× Few candidates explained the cooling effect of air ventilation design of the buildings.

× Candidates failed to consider the socio-economic cost brought by the adverse impact of the heat island effect.

× They commonly put forward one-sided, superficial arguments in evaluating the cost-effectiveness of the measure only, such as 'increase in construction or maintenance costs' or 'unsafe' etc.

× Few candidates discussed the feasibility of adopting the measure with reference to the specific urban environment of Hong Kong.

Section C Question 6

Area	Key concepts/ skills
Natural hazards (earthquakes, volcanic eruptions) commonly found in areas with active tectonic activities (divergent plate boundaries)	• Plate movements along divergent plate boundaries • The relationship between plate movements and natural hazards • Occurrence of earthquakes, volcanic eruptions • Systematic and logical explanation
Tectonic activities and their resultant natural hazards at divergent and convergent plate boundaries. Factors affecting the impacts of natural hazards	• Spatial distribution of divergent and convergent plate boundaries • Geographical setting (physical and human environment) of divergent and convergent plate boundaries • Man - environment interrelationship: natural hazards and socio-economic loss • Factors affecting the magnitude and intensity of natural hazards: location, nature, scale etc. • Making comparison • Systematic and logical presentation of ideas

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Section C Question 6 (Cont')

Account for the occurrence of earthquakes and volcanic eruptions along divergent plate boundaries.

- ✓ Most candidates demonstrated a basic understanding of plate tectonics and the occurrence of natural hazards.
- ✓ Candidates commonly described accurately the plate movements along divergent plate boundaries.
- ✓ Some candidates cited correct examples of divergent plate boundaries.

- ✓ Compared with volcanic eruption, candidates explained more systematically the occurrence of earthquakes and in greater detail.
- ✓ A few candidates explained accurately how the friction between plates leads to a build-up of stress in the rock.

- × Few candidates explained specifically the occurrence of earthquakes with reference to the transform faults formed as plates move apart.

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Section C Question 6 (Cont')

Account for the occurrence of earthquakes and volcanic eruptions along divergent plate boundaries.

- × Candidates commonly provided a brief account of the occurrence of volcanic eruption in general.
- × Only a few candidates explained specifically the release in pressure at divergent plate boundaries causing rocks underneath to melt into magma.
- × Quite a number of candidates named some volcanic islands as examples but they commonly failed to use them to illustrate the occurrence of volcanic eruption. Some examples given were irrelevant such as Hawaii islands.

Explain why the earthquakes and volcanic eruptions that occur along divergent plate boundaries usually bring less socio-economic losses to the local areas when compared with those that occur along convergent plate boundaries.

- × In the second part of the question, most candidates failed to answer satisfactorily.
- × Some candidates misinterpreted it as the differences between more developed with less developed regions.

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Section C Question 6 (Cont')

Explain why the earthquakes and volcanic eruptions that occur along divergent plate boundaries usually bring less socio-economic losses to the local areas when compared with those that occur along convergent plate boundaries.

× Candidates apparently did not realise that spatial distribution and location of plate boundaries are the key factors affecting the population exposed and vulnerable to the hazards.

× Only a few candidates compared the physical and human settings of the hazard-prone areas at the two types of plate boundaries, and explained how they cause the differences in socio-economic losses.

× Many candidates explained with reference to the severity of hazards. However, they did not demonstrate adequate understanding of the differences in the occurrence and the characteristics of the hazards along the two types of plate boundaries. Thus, the explanations were commonly brief and general.

× Many candidates provided irrelevant answers, such as the occurrence of natural hazards along convergent plate boundaries or secondary hazards etc.

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Section C Question 7

Area	Key concepts/ skills
Factors affecting the location of IT industry in US	• Location of IT headquarters in US e.g. Silicon Valley etc. • Nature of IT industry: science and technology based • Major factors affecting the location of IT industry: research and development, labour quality, agglomeration etc. • Systematic and logical explanation
Impacts of globalization and technological advances on the mode of production and location of IT plants	• Multi-point and transnational production, and their merits • Change of location of IT plants over time • Effects of globalization and technological advances on the location of IT plants • Systematic and logical presentation of ideas

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Section C Question 7 (Cont')

Account for the locational factors of the headquarters of the IT industry in the US.

✓ In the first part of the question, most candidates identified the general location factors of R&D centres of IT industries, such as proximity to universities, high accessibility, industrial agglomeration and pleasant environment etc.

× Many candidates' explanations were too general or outdated as they were insufficient to account for the contemporary location of IT industry in the US.

× Few candidates pointed out specifically the importance of innovative culture and atmosphere that attracts professional talents or the advantages of branding and business linkages brought by agglomeration.

× Candidates commonly did not understand the other functions, in addition to R&D, of the headquarters. Thus, they failed to account for the specific location factors, such as availability of financial capital and management personnel, and strong government etc.

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Section C Question 7 (Cont')

Account for the locational factors of the headquarters of the IT industry in the US.

- × Most candidates cited Silicon Valley as an example. However, many failed to explain with reference to its geographical setting and development, probably due to their inadequate knowledge and understanding.
- × Quite a number of candidates failed to write the name 'Silicon' Valley correctly in Chinese.

Why does the IT industry in the US at present commonly adopt the mode of multi-point production?

- × In the second part of the question, most candidates' understanding of multipoint production was superficial in that it was limited only to R&D in MDCs, and production in LDCs in order to cut down cost.
- × They did not understand the global marketing strategy, specialisation and division of work involved in multipoint production. Thus, they failed to account for the setting up of global parts production plants, regional R&D centres, and regional headquarters by US IT corporations.

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Section C Question 7 (Cont')

Why does the IT industry in the US at present commonly adopt the mode of multi-point production?

× Candidates commonly failed to explain accurately the role of technological advancement in logistics and telecommunications in building up a reliable and efficient global supply chain, which is essential for multi-point production.

× Candidates commonly failed to focus on the characteristics of IT industries, including labour intensive in production, 'short product life cycle' requiring high flexibility in the scale of production etc., to account for the merits of outsourcing production to overseas.

× Many candidates did not realise that IT industries are relatively clean, and inaccurately suggested 'stricter environmental control in the US' as a factor.

× Many candidates described the advantages of setting up plants in China and India etc. which were irrelevant.

× Quite a number of candidates gave unsound explanations, such as 'to provide job opportunities and to improve the living of LDCs people', or 'to shift pollution problems to LDCs' etc.

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Section C Question 8

Area	Key concepts/ skills
Physical, economic, social and political factors affecting agricultural production in Sahel	• Location of Sahel • People – environment interrelationship • Characteristics of the physical environment of Sahel: climate, soil, vegetation cover, relief etc. and their influences on agriculture • Characteristics of the human environment of Sahel: technological, economic, political and socio-cultural aspects and their influences on agriculture • Agricultural characteristics of farming in Sahel e.g. nomadic herding • Systematic and logical explanation
Measures of sustainable agricultural development	• People – environment interrelationship • Change over time • Sustainable agricultural development • Measures of sustainable agricultural development: -management of agricultural resources (land, water etc.), use of appropriate technology etc. • Systematic and logical presentation of ideas

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Section C Question 8 (Cont')

Account for the physical and human factors of low food production in Sahel.

✓ In the first part of the question, most candidates described the general physical settings of Sahel. Some candidates wrote lengthy and detail descriptions of the region's environment.

× Some candidates confused seasonal rainfall with unstable or unreliable rainfall.
× Some others inaccurately described Sahel as an arid region.

× Most candidates' explanations for low food production in Sahel were brief and general only.
× They were unable to explain specifically how the length of growing season, the area of irrigated farmland, the intensity of farming, and the output per hectare of farmland etc. are affected.

× Many candidates inaccurately suggested high evaporation rate as the cause of low rainfall, low relative humidity or even infertile soil.

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Section C Question 8 (Cont')

Account for the physical and human factors of low food production in Sahel.

✓ Candidates commonly described the general socio-economic causes of famine in LDCs, such as lack of capital, low education level, corrupted government and civil war etc.

× Only a few candidates were able to account for low food production with reference to the specific conditions in Sahel, such as nomadic herding, population growth and governments' agricultural policies etc.

× Some candidates described the routine farming practice of nomadic herding which was irrelevant.

× Some others wrongly described farming in Sahel as cattle rearing.

Explain how local people may change their farming methods to produce enough food in the long run.

× In the second part of the question, candidates commonly performed poorly.

× Most candidates failed to respond to the main points of 'local people', 'farming methods' and 'in the long run' stated in the question when giving answers.

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Section C Question 8 (Cont')

Explain how local people may change their farming methods to produce enough food in the long run.

- × Many candidates proposed expensive and high-tech precision farming or growing of GM crops that local people cannot afford.
- × Some candidates proposed modern farming methods e.g. using chemicals or machines that are not sustainable.

- × Candidates commonly showed inadequate understanding of sustainable farming.
- × Some candidates mentioned farming methods such as intercropping, crop rotation, dry farming and mulching etc. However, they either gave brief and general explanations, or inaccurately focused on maintaining soil fertility.
- × Some candidates suggested transhumance. They apparently did not realise that this existing farming method is unable to cope with increasing population.

- × Some others suggested measures that are not 'farming methods', such as seeking international aids, constructing dams and aqueducts, providing education and birth control etc.

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Q&A

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2022 HKDSE Geography Examination

Review of Candidates' Performance in Paper 2

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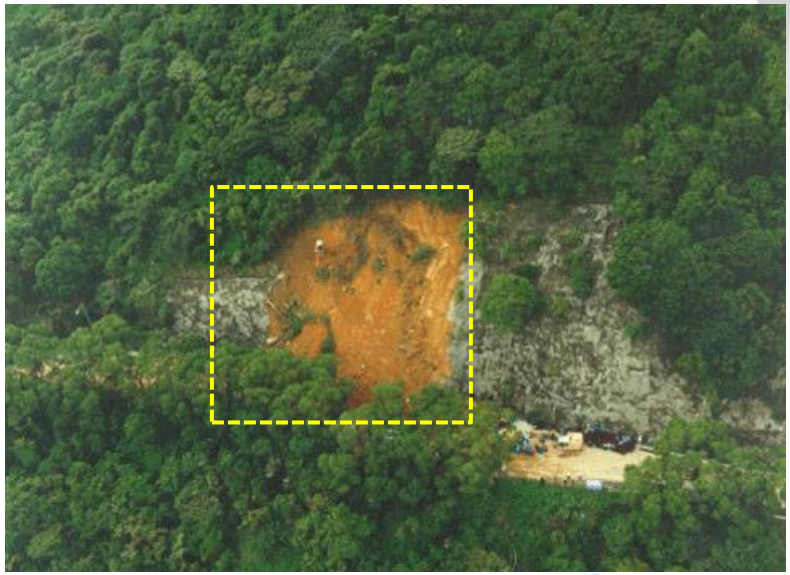
Section D: Question 1

1. Elective: Dynamic Earth

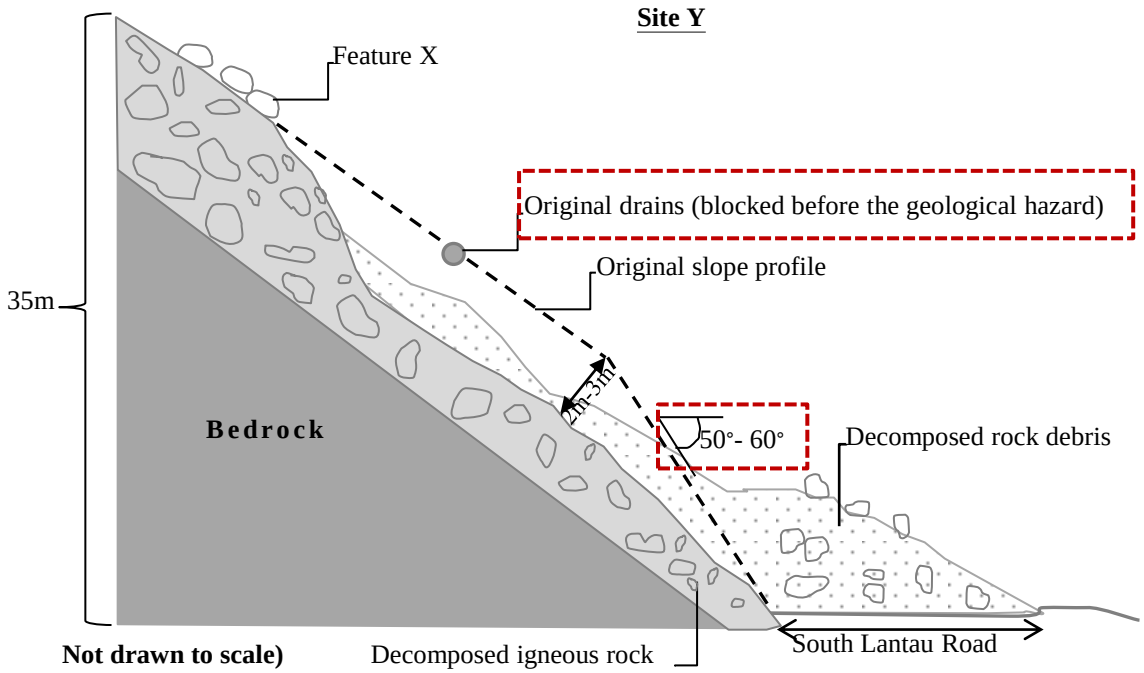
Figure 1a shows feature X and some information of a geological hazard occurred in August in a particular year at site Y along South Lantau Road. Photograph 1b shows a slope management strategy at another site along South Lantau Road.

Figure 1a



Section D: Question 1 (Cont')

Figure 1a



Beginning time of the rainstorm	2 days before the geological hazard
Rainfall amountandrainfall **duration**	692mm, 48 hours
Remarks	The intensity of rainfall was recorded as occurring about once every 150 years.

Section D: Question 1 (Cont')

(a)	**Refer to Figure 1a.**	
(i)	Identify feature X.	(1 mark)
(ii)	With evidence from the figure, explain how **erosion** may lead to the formation of feature X.	(4 marks)
(b)	**Refer to Figure 1a.**	
(i)	With reference to the **physical** conditions at site Y, account for the occurrence of the geological hazard.	(5 marks)
(ii)	Explain how **human** factors would lead to the occurrence of this geological hazard.	(4 marks)

Section D: Question 1 (Cont')

(c)	**Refer to Figure 1a and Photograph 1b.** Discuss whether the slope management strategy is an appropriate measure to enhance slope safety **at site Y.**	(4 marks)

Photograph 1b

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Section D: Question 1 (Cont')

Curriculum Content – Dynamic Earth

Part (a):

- Geomorphology and geology of Hong Kong
  - Rock types and their distribution in Hong Kong
  - External processes (weathering, erosion, mass wasting)

Parts (b) & (c):

- Geological resources - Landslides
  - Causes of landslides in Hong Kong, including natural and human factors
  - Slope management and landslide prevention

Section D: Question 1 (Cont')

What Most Candidates Knew:

Part (a):

- Identifying the 'tor'
- The process of spheroidal weathering

Part (b):

- 'heavy rain' and 'steep slope' as physical factors read from question data
- 'blockage of drains' as human factor read from question data

Part (c):

- identifying the 'flexible barrier' and its function to hold back landslides
- citing that it is a means of protection but not prevention

Section D: Question 1 (Cont')

Common Mistakes/ Misconceptions of Candidates:

- Lengthy account of the processes of 'weathering' instead of 'erosion' that lead to formation of feature X
- Misusing the rainstorm data to explain the formation of feature X *(Rome is not built in a day)*
- Wrongly citing 'removal of vegetation', 'blocking of drainage' etc. as phyiscal conditions leading to occurrence of landslide
- Reciting irrelevant factors e.g. 'slope filling', 'building of squatters' from textbooks or far-fetched inferences e.g. 'heavy road traffic', ' removal of vegetatyion' as human factors
- Making vague statements e.g. 'cheap to build', 'easily damaged by landslides' in discussing the slope management strategy
- Wrongly identifying the 'maintenance access' as a strategy

Section D: Question 1 (Cont')

Directions for Better Performance:

- ***Make close reference to the question wording or data provided to answer using the relevant concepts***

(a)	**Refer to Figure 1a.**
(ii)	With evidence from the figure, explain how **erosion** may lead to the formation of feature X.
(b)	**Refer to Figure 1a.**
(ii)	Explain how **human factors** would lead to the occurrence of this geological hazard.

- ***Discuss the application of the slope management strategy at the 'actual setting of the site'***

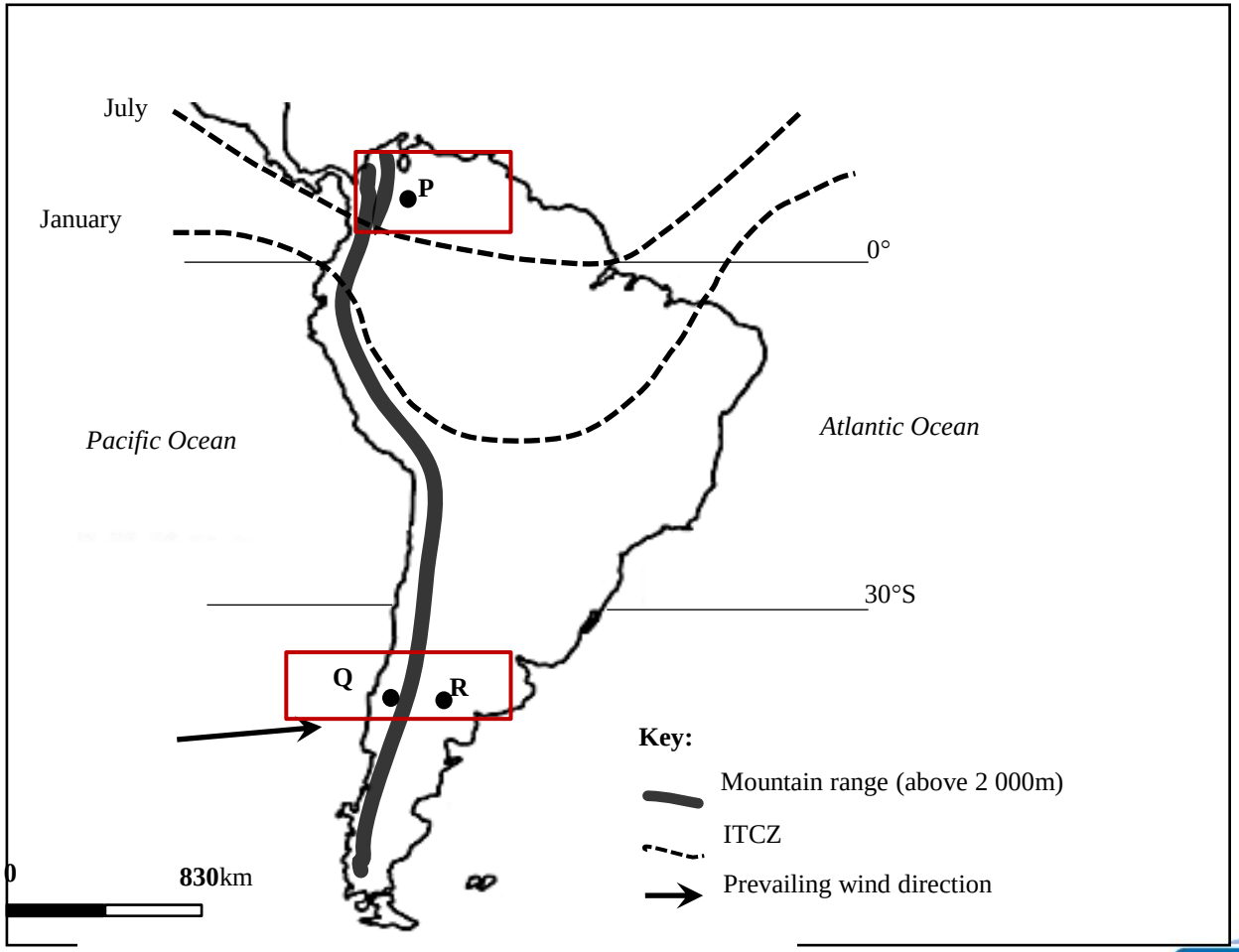
Section D: Question 2

2. Elective: Weather and Climate

Figure 2a shows the distribution of the major mountain range in South America. It also shows the locations of Intertropical Convergence Zone (ITCZ) in January and July and the locations of areas P, Q and R. Figure 2b shows the climatic conditions of area P. Table 2c shows some information of areas P, Q and R.

Section D: Question 2 (Cont')

Figure 2a



Section D: Question 2 (Cont')

Figure 2b

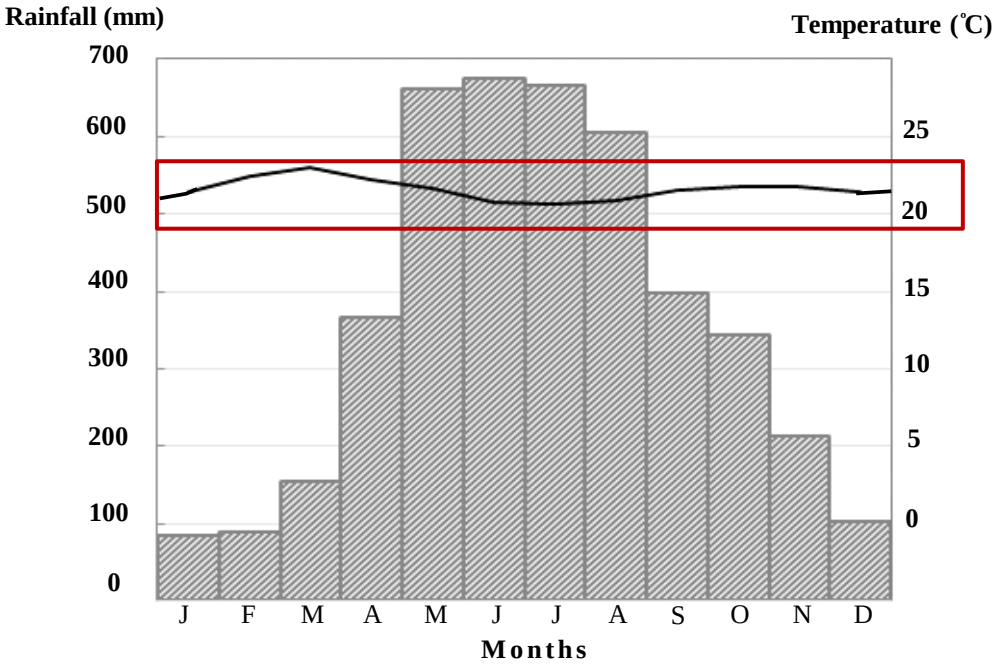


Table 2c

	Area P	Area Q	Area R
Altitude (m)	226	171	241
Annual rainfall (mm)	4338	3 056	277

Section D: Question 2 (Cont')

(a)	**Refer to Figure 2a.**	
(i)	Describe the **locations** of the ITCZ in January and July.	(2 marks)
(ii)	Explain the occurrence of the ITCZ at the above locations.	(4 marks)
(b)	**Refer to Figure 2a, Figure 2b and Table 2c.**	
(i)	Describe the rainfall pattern of area P.	(2 marks)
(ii)	Explain how the ITCZ affects the rainfall pattern of area P.	(4 marks)
(c)	**Refer to Figure 2a and Table 2c.**	
(i)	Explain the relationship between **relief and annual rainfall** of area R.	(2 marks)
(ii)	Area Q and area R are located at the same latitude, but there is significant difference in annual rainfall. Discuss the relative importance of **relief** in affecting the annual rainfall of area Q.	(4 marks)

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Section D: Question 2 (Cont')

Curriculum Content – Weather and Climate

Part (a):

- Atmospheric motion
  - The tri-cellular model
  - Global pressure pattern

Part (b) & (c):

- Moisture in the atmosphere
  - Humidity and condensation
  - Types of precipitation
  - Global precipitation distribution pattern

Section D: Question 2 (Cont')

What Most Candidates Knew:

Part (a):

- Description of ITCZ locations with reference to latitudinal positions
- Relating seasonal positiion of 'overhead sun' to changing location of ITCZ

Part (b):

- Describing the rainfall condition as 'high' or identifying the months of higher and lower rainfall
- Identifying the relationship between ITCZ location and rainfall pattern of area P

Part (c):

- The process of 'relief rain' formation

Section D: Question 2 (Cont')

Common Mistakes/ Misconceptions of Candidates:

- Not pointing out the latitudinal range of ITCZ locations correctly
- Not pointing out the 'overhead sun' positions at winter solstice and summer solstice respectively
- Wrongly stating the convergence of trade winds as primary force for ITCZ formation
- Wrongly describing the rainfall pattern as 'in summer /winter' in the hot all year climate of area P
- Wrongly relate the rainy season of area P to position of 'overhead sun'
- Unable to recognize the 2 000m mountain as relief barrier causing lower rainfall at area R
- Wrongly citing 'altitude' as 'relief'
- Overlooking 'winds' and 'moisture in air' as the other factors affecting relief effect on rainfall

Section D: Question 2 (Cont')

Directions for Better Performance:

- ***Do more practice in reading and interpretation of different types of maps***
- ***Do more practice in reading climatic data / climatic graphs***
- ***Improve understanding of the relationship between 'overhead sun', 'ITCZ formation', 'trade wind convergence' and rainfall conditions***
- ***Improve data interpretation skills e.g. 'mountain range' in Figure 2a, 'altitude' in Table 2c.***
- ***Applying knowledge of winds / airmasses, moisture in the air, relief effect (causing uplifting, expansion, cooling and condensation) in the discussion***

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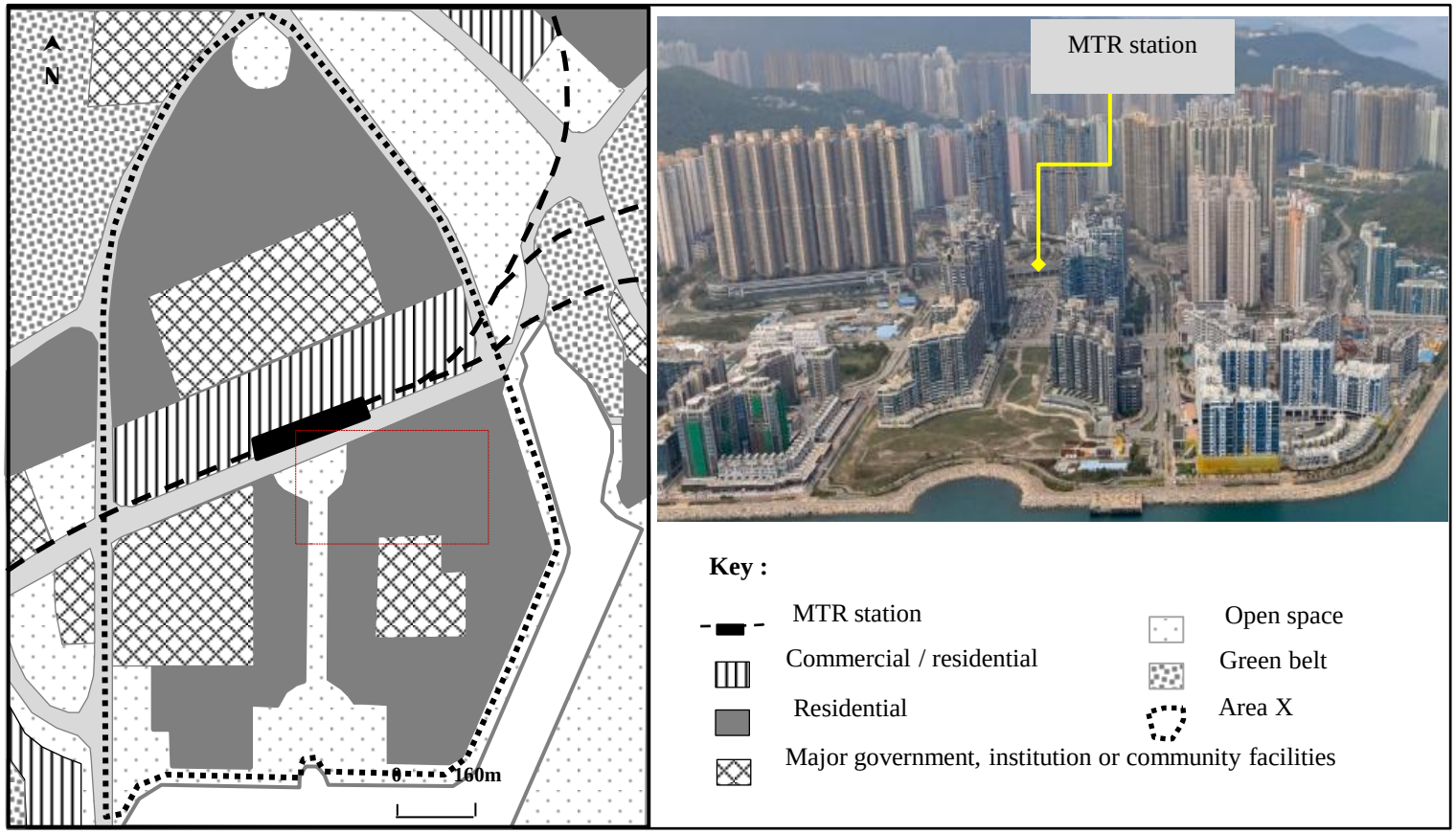
Section D: Question 3

3. Elective: Transport Development, Planning and Management

Figure 3a shows a sketch map and a photograph of the land use near the MTR Tseung Kwan O Station. Figure 3b shows some information of major transport infrastructure in Tseung Kwan O New Town. Table 3c and Table 3d show some population information of Tseung Kwan O New Town.

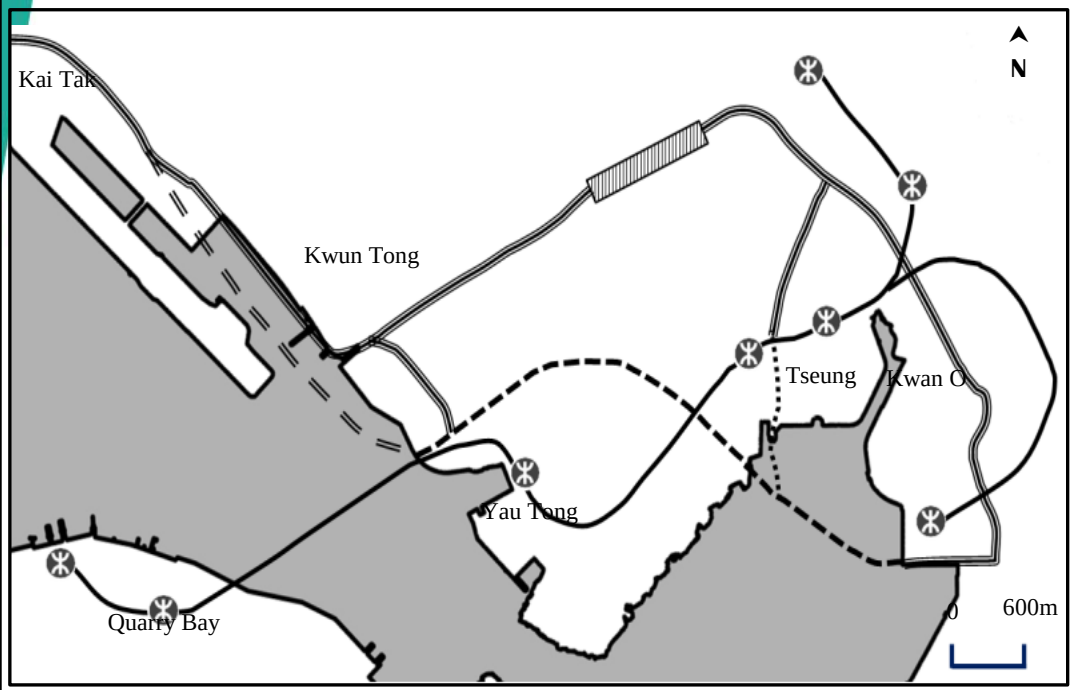
Section D: Question 3 (Cont')

Figure 3a



Section D: Question 3 (Cont')

Figure 3b



Key:

- MTR Tseung Kwan O Line
- Tseung Kwan O Tunnel
- Tseung Kwan O - Lam Tin Tunneland Cross Bay Link (under construction)
- Connecting road (under construction)
- Major road (existing)
- Major road (proposed)

Tseung Kwan O Tunnel(opened in 1990)	**Design capacity**(number of **vehicles** per day)			78500		
	Average daily vehicular flow	Number of vehicles		72 200	67 700	93500
		% of types of vehicles	**Private cars, taxis and motor cycles**	59.1	63.9	75.8
			Buses and others	40.9	36.1	24.2
MTR Tseung Kwan O Line (opened in 2002)	**Loading** rate* (%)*(*4 persons standing per sq. metre)*			No data		100

Section D: Question 3 (Cont')

Table 3c

Population of Tseung Kwan O New Town	
Year	Population
2004	331 000
2018	400 000
2029*	451 600
*Projected figures	

Table 3d

Workplace of the working population in Tseung Kwan O New Town (2016)	
Workplace	Working Population(%)
Tseung Kwan O	9.7
Other districts in Hong Kong	71.0
No fixed place/ Work at home	19.3

(a)	**Refer to Figure 3a.**	
(i)	Describe the **land use characteristics** of area X shown in the sketch map.	(3 marks)
(ii)	Explain how the location of the MTR station has influenced the land use characteristics mentioned in (a) (i).	(3 marks)

Section D: Question 3 (Cont')

(b) (i)	**Refer to Figure 3b.** Describe and explain the impact brought about by the MTR Tseung Kwan O Line on the vehicular flow of the Tseung Kwan O Tunnel in 2004.	(4 marks)
(ii)	**Refer to Figure 3b, Table 3c and Table 3d.** Describe the changes in the vehicular flow of the Tseung Kwan O Tunnel from 2004 to 2018. Explain the related transport problem.	(4 marks)
(c)	**Refer to Figure 3b, Table 3c and Table 3d.** Discuss whether the construction of Tseung Kwan O - Lam Tin Tunnel and Cross Bay Link to improve the transport network is a **sustainable transport development**.	(4 marks)

Section D: Question 3 (Cont')

Curriculum Content – Transport Development, Planning and Management

Part (a):

- Impact of transport on urban land use
- The impact of "transit-oriented development" on the transport and urban development in Hong Kong

Part (b):

- Major transport modes in Hong Kong
- Improvement of transport infrastructure in Hong Kong
- Expansion and improvement of public transport in Hong Kong

Part (c):

- Various measures involved in managing road use in Hong Kong
- Development of a sustainable transport system in Hong Kong

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Section D: Question 3 (Cont')

What Most Candidates Knew:

Part (b):

- Correct description of the drop in number of vehicles using the Tsueng Kwan O Tunnel brought about by MTR TKO Line from the question data
- Correct description of the increase vehicular flow of the Tsueng Kwan O Tunnel from 2004 to 2018, leading to traffic congestion

Part (c):

- Citing that the new transport project helped divert some traffic from Tseung Kwan O Tunnel to relieve the congestion problem

Section D: Question 3 (Cont')

Common Mistakes/ Misconceptions of Candidates:

- Unable to identify the central location of the MTR station in describing the land use characteristics
- Copying the types of land uses from the map key instead of describing their characteristics
- Showing no knowledge of the concept of 'transit-oriented development'
- Showing limited knowledge of merits of MTR other than that it was faster and more convenient
- Wrong interpretation of the working population in TKO data
- Concept of sustainability of transport development not well understood - unable to discuss impact of the new transport project holistically

Section D: Question 3 (Cont')

Directions for Better Performance:

- ***Do more practice in reading and interpreting land use maps and photos***
- ***Improve understanding of the 'spatial' concept***
- ***Improve understanding of the 'transit-oriented development' concept***
- ***Grasp the important concepts in transport geography, e.g. accessibility, connectivity***
- ***Improve understanding of the characteristics of different modes of transport and the relationship between population and the demand for transport***
- ***Better understanding of the concept of 'sustainability and development of transport systems'***

Section D: Question 4

4. Elective: Regional Study of Zhujiang (Pearl River) Delta

Table 4a shows some information of the Zhujiang Delta Region in 2000 and 2019. Table 4b shows some information of Shenzhen in 2000 and 2019. Figure 4c shows the concentration of some air pollutants in Shenzhen in 2000, 2010 and 2019.

Table 4a

		2000	2019
Urbanisation (%)		71.6	86.3
Employment structure (%)	Primary industry	33.9	1.6
	Secondary industry	32.0	41.3
	Tertiary industry	34.1	57.1

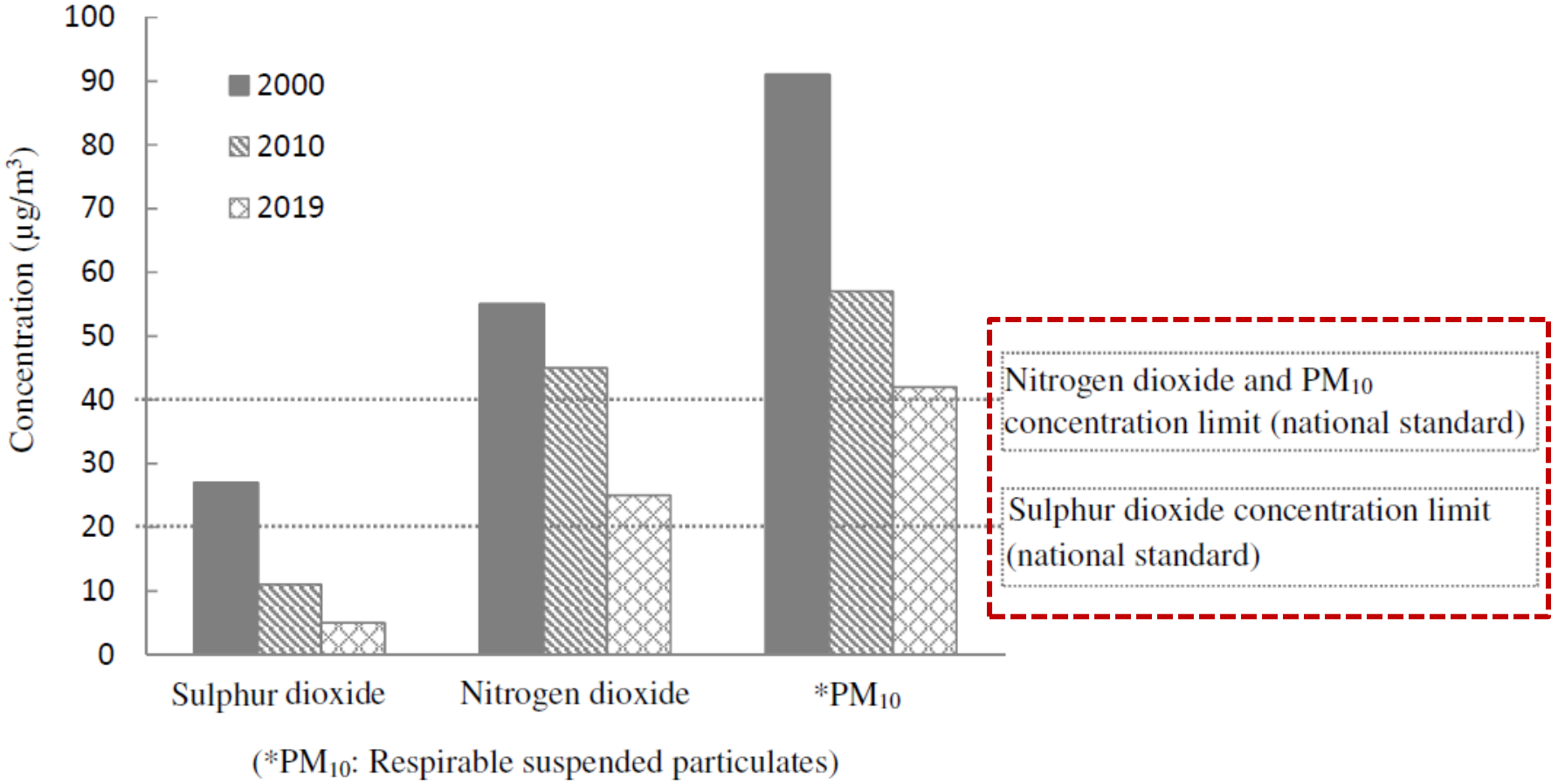
Section D: Question 4 (Cont')

Table 4b

		2000	2019
Population ('0 000)		701.24	1 343.88
Urbanisation (%)		92.46	99.52
Employment structure (%)	Primary industry	1.3	0.001
	Secondary industry	57.0	39.7
	Tertiary industry	41.7	60.2
Value of economic output (hundred million RMB)	Primary industry	22.19	26.93
	Secondary industry	1 165.08	10 501.57
	Tertiary industry	1 031.93	16 398.6
Types of industry with higher value (percentage share in total industrial output)	Electronic and communication equipment manufacturing	44.3	59.9
	Electrical machinery and equipment manufacturing	6.7	7.8
	Rubber and plastic products	4.5	3.1
	Metal products	4.0	1.6
	Textile, garment and related products manufacturing	2.2	0.98

Section D: Question 4 (Cont')

Figure 4c



(*PM_{10}: Respirable suspended particulates)

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Section D: Question 4 (cont.)

(a)	**Refer to Figure 4a.** Describe and explain the **relationship between the changes** in urbanisation level and employment structure in the Zhujiang Delta Region in 2000 and 2019.	(5 marks)
(b)	**Refer to Table 4a and Table 4b.** The changes in the employment structure in Shenzhen were different from the overall changes in the Zhujiang Delta Region in 2000 and 2019.	
(i)	Describe these differences.	(2 marks)
(ii)	With evidence from Table 4b, explain the changes in the employment structure of secondary industry and tertiary industry in Shenzhen.	(5 marks)

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Section D: Question 4 (Cont')

(c)	**Refer to Table 4b and Figure 4c.**	
(i)	With reference to the **national standards** shown in Figure 4c, describe the changes in the concentration of different air pollutants in Shenzhen from 2000 to 2019.	(2 marks)
(ii)	With reference to the percentage changes in the total industrial output in 2000 and 2019 as shown in Table 4b, discuss the **relative importance of these changes** in affecting the air pollutant concentration in Shenzhen.	(4 marks)

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Section D: Question 4 (Cont')

Curriculum Content – Regional Study of Zhujiang Delta

Parts (a) & (b):

- Changing land use pattern since the 1980s: from rural-agricultural dominant to urban-industrial dominant
- Changes and development in manufacturing industries of Zhujiang Delta

Part (c):

- Environmental pollution in Zhujiang Delta: causes, types, variation over time
- Management strategies

Section D: Question 4 (Cont')

What Most Candidates Knew:

Part (a):

- Identification of 'positive' and 'negative' relationships

Part (b):

- Correct description of the different changes in secondary and tertiary employment between Shenzhen and ZDR

Part (c):

- Correct description of the changes in concentration of different pollutants in Shenzhen over the years
- Citing question data of changes in industrial employments in the discussion

Section D: Question 4 (Cont')

Common Mistakes/ Misconceptions of Candidates:

- Unable to explain the relationship between changes in urbanisation level and employment structure in ZDR
- Unable to identify the 'changes in employment structure of the two places', or the 'differences' between the two
- Citing irrelevant material e.g. 'Emptying the cage and let the right birds in' policy instead of using the question data
- Unable to make proper use of question data to explain the changes in employment structure of Shenzhen
- 'Electronic and communication equipment' wrongly quoted as 'tertiary industry'
- Not referring to the 'national standard' of different pollutants
- Unable to consider both positive and negative effect of 'other factors' in discussing the relative importance
- Mixing up air pollutants with greenhouse gases

Section D: Question 4 (Cont')

Directions for Better Performance:

- *Better understanding of key terms: 'relationship', 'changes between years', 'differences between places' in answering*
- *Comparing the 'rate of change' in addition to 'magnitude of change' in describing changes over time*
- *Making proper use of question data by interpretation instead of direct copying in explanations*
- *Considering 'other factors' as well as their possible 'positive' and 'negative' influences in the discussion*

Section E: Question 5

5. Elective: Dynamic Earth

Describe the characteristics of siltstone found in Hong Kong and explain its formation process. Comment on the significance of sedimentary rock in shaping the physical landscape of Hong Kong. (12 marks)

Curriculum Content – Dynamic Earth

- Geomorphology and geology of Hong Kongh
  - Overall landform distribution
  - Rock types and their distribution in Hong Kong

Section E: Question 5 (Cont')

What Most Candidates Knew:

- General description of characteristics of clastic sedimentary rocks
- Simple knowledge of siltstone
- Sedimentary rocks are found in NE Territories
- General description of processes of sedimentary rock / siltstone formation
- Citing classical example of Pat Sin Leng conglomerate and Ping Chau siltstone
- Some erosional landforms in Hong Kong
- How various types of rocks and other factors shape the physical landscapes in Hong Kong

Section E: Question 5 (cont.)

Common Mistakes/ Misconceptions of Candidates:

- Lacking elaboration of siltstone characteristics e.g. 'less resistant'
- Listing geological terms related to processes of siltstone formation but not providing explanation
- Lengthy account of processes of weathering and erosion leading to siltstone formation
- Unable to name the exact location of siltstone as the local example
- Unable to match the exact place of landforms mentioned with the type of sedimentary rock and its characteristics (e.g. waterfall in Bride's Pool - differential weathering of conglomerate and sandstone)
- Providing brief account of sedimentary rocks but lengthy account of other rocks in shaping the physical landscape in Hong Kong

Section E: Question 5 (Cont')

Directions for Better Performance:

- Compare different rock types in learning their characteristics
- Better understanding of the processes of siltstone formation
- Learning the distribution of different types of sedimentary rocks in Hong Kong
- Learning more examples of landforms in Hong Kong related to different types of sedimentary rocks
- Better understanding of how some characteristics (e.g. resistance to denudation, bedding planes, angle of dip) of sedimentary rocks shape the physical landscape

Section E: Question 6

6. Elective: Weather and Climate

Describe the conditions favouring the formation of typhoons and explain their formation process. Discuss the **relationship** between monsoon winds and the development of typhoons in the South China region. (12 marks)

Curriculum Content – Weather and Climate

- Factors affecting the climate of a location
- An introduction of the climate and major climatic zones of China
- Seasonal occurence of weather systems in Hong Kong and the Zhujiang Delta Region: formation and impact - typhoons

Section E: Question 6 (Cont')

What Most Candidates Knew:

- Brief description of conditions favouring typhoon formation
- Brief explanation of processes of typhoon formation
- General structure of typhoons
- Characteristics and formation of monsoon winds
- Brief account of impact summer monsoons (provide moisture) and winter monsoons (cold and dry) on typhoon development in South China region

Section E: Question 6 (Cont')

Common Mistakes/ Misconceptions of Candidates:

- Inaccurate knowledge of the sea water temperature, latitudinal range etc. for typhoon formation
- Poor understanding of effect of 'Coriolis force' and 'latent heat' on typhoon formation
- Listing other factors e.g. vertical wind shear without understanding
- Describing the structure of typhoon instead of explaining its formation process
- Unable to discuss the 'two-way' relationship between monsoon winds and typhoon development

Section E: Question 6 (Cont')

Directions for Better Performance:

- Better understanding of the conditions favouring typhoon formation
- Better understanding of the processes involved e.g. latent heat, Coriolis force
- Learning the processes involved in the stages of typhoon development in explaining its formation
- Discuss the 'two-way' relationship between monsoon winds and typhoon development
- Paying more attention to daily weather conditions in learning this topic

Section E: Question 7

7. Elective: Transport Development, Planning and Management

Describe the factors influencing the modal choice of sea and air freight tansport. Explain how **multi-modal** freight transport is adopted in Hong Kong. (12 marks)

Curriculum Content – Transport Development, Planning and Management

- Transport patterns (freight transport)
- Distance, transport cost, modal choice and modal competition
- Unimodal transport and multimodal transport

Section E: Question 7 (Cont')

What Most Candidates Knew:

- The main difference beween sea and air freight transport - cost, speed of delivery, nature of cargo
- Simple example of goods carried
- Different modes of transport and main transport infrastructures in Hong Kong
- Need for transfer of goods between modes of transport

Section E: Question 7 (Cont')

Common Mistakes/ Misconceptions of Candidates:

- Unable to compare the 'terminal and hauling costs'of air and sea freight transport
- Not providing adequate examples to illustrate the different characteristics between air and sea freight transport
- Mixing up freight and passenger transport in affecting the modal choice
- Mistaking 'multi-modal' transport as various modes of transport
- Limited understanding of Hong Kong as a transport and logistic hub, leading to the large demand for transhipment of goods between different transport modes
- Lacking the knowledge that multi-modal transport is a common practice used by transport providers

Section E: Question 7 (Cont')

Directions for Better Performance:

- Better understanding of the concept of 'Hong Kong as a transport and logistics hub'
- Distinguishing between passenger and freight modes of transport
- Learning the characteristics of different transport modes in greater detail, with relevant examples of cargo types
- Better understanding of the characteristics of 'multi-modal' transport and the contributing factors in Hong Kong

Section E: Question 8

8. Elective: Regional Study of Zhujiang (Pearl River) Delta

Describe the changes in agriculture in the Zhujiang Delta Region since the 1980s. Discuss the **relative importance** of the development of agricultural technology in **affecting these changes**. (12 marks)

Curriculum Content – Regional Study of Zhujiang Delta

- Changes and development in agriculture of Zhujiang Delta - from subsistence to export-oriented and internal consumption
- Reasons for such changes and development

Section E: Question 8 (Cont')

What Most Candidates Knew:

- Recent changes of agriculture in the Zhujiang Delta Region
- Citing the key concepts related to the change: from 'subsistence to export-oriented and internal consumption', 'intensification', 'specialisation' etc.
- The factors leading to the changes (NOT REQUIRED in this question)
- Various forms of modern farming technology
- Explaining how the farming technology helps increase output / efficiency

Section E: Question 8 (Cont')

Common Mistakes/ Misconceptions of Candidates:

- Unable to elaborate well on the changes in agriculture mentioned
- Unable to provide adequate relevant examples to illustrate the changes mentioned
- Writing lengthy answers on 'why' instead of 'how' the changes occurred
- Unable to understand that the various changes occurred at different time periods since the 1980s
- Not relating the importance of various forms of farming technology to the changes mentioned in the first part of the question
- Unable to discuss the relative importance of farming technology by looking into other factors affecting the changes in agriculture

Section E: Question 8 (Cont')

Directions for Better Performance:

- Better understanding that the various changes of agriculture in the Zhujiang Delta Region since 1980s occur in different time periods
- Better understanding of how the various changes occurred, and learning from relevant examples
- Discussion of the importance of farming technology should be based on the changes mentioned in the first part of the question
- Discussion of the 'relative' importance of farming technology should consider that of other factors e.g. government policy, urban and industrial development

General Comments and Recommendations

1. *Candidates should improve their knowledge of **key geographical concepts**, and apply them correctly in answering relevant topics.*
2. *Candidates should read into the theme and key terms of the questions carefully to avoid reproducing lengthy insignificant or irrelevant materials learnt by rote. **Relevance and quality** rather than volume of written materials will help their performance.*
3. *Candidates should learn to **compare relationships between data** properly. They should also practise describing and explaining the trend and **changes** of the data correctly.*

General Comments and Recommendations (Cont')

4. *Candidates should strengthen their understanding of locational and **spatial concepts**, with more practice in **reading and interpreting different types of maps**.*
5. *In answering data / skill-based questions, candidates should learn to **select and interpret the data** provided, rather than copying information without analysing it.*
6. *Candidates should **update** their geographical **knowledge and information** when studying various modules, for example from revising the **past examination papers** and by learning from their **daily experiences**.*
7. *In answering questions that require discussions, candidates' arguments should be **based on the scenario** given in the questions. Their statements should be logical, concrete and from **multi-perspectives**.*

Q&A

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HKDSE Geography Assessment Frameworks

2023 Streamlining

	Paper 1	Paper 2
Weighting:	75%	25%
Duration:	~~2¾~~ 2½ hours	1¼ hour
Scope:	Compulsory Part	Elective Part
Section/ Question type:	A. Multiple-choice (20% → 24%) B. Fieldwork-based & Data/Skill-based/ Structured (15% → 39%) ~~C. Data/Skill-based/ Structure (20%)~~ ~~D~~ C.Short Essay (10% → 12%)	~~E~~ D.Data/Skill-based/ Structured (15%) ~~F~~ E.Short Essay (10%)

For the 2023 HKDSE, candidates are expected to have fieldwork experiences in the following ~~modules~~ module:

- ~~Opportunities and Risks~~
- ~~Managing River Environment~~
- Disappearing Green Canopy

For further details , please go to

https://www.hkeaa.edu.hk/DocLibrary/HKDSE/Subject_Information/geog/2023hkdse-e-geog-rev1.pdf

Reminders for Candidates in 2023 Geography Examination

1. ***DO NOT*** *allow to use self-made transparent plastic sheets with gridlines in the examination*
2. ***DO NOT*** *use colour pencils, colour marker pens or highlighters*
3. ***CLEAR HANDWRITING*** *and* ***PAGE LAYOUT*** *are essential for accurate marking*
4. ***WRITE CLEARLY*** *by crossing out/ deleting all unwanted words*
5. *Mark carefully and accurately the* ***QUESTION NUMBER*** *on* ***EACH PAGE*** *of the Answer Book, especially in* ***Sections B & C of Paper 1***
6. *Start* ***EACH*** *question (not part of a question) on a* ***NEW*** *page*
7. *Study carefully the 2023 Handbook for Candidates*

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Thank You for Your Participation

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